

Dear study participant, thank you for your interest and commitment to this study on making decisions with visualizations.

Aim of the Study

This research is being undertaken by the University of Rostock to understand how people use data visualizations to make decisions. A decision is about finding the most suitable option among a number of alternatives, like choosing the best hotel for a holiday trip or the best apartment to rent. You will see different visualizations and you will be asked to make a decision based on the visualized data.

Study Content

Imagine you are planning a holiday and want to book a hotel. The search engine lists 100 hotels to choose from. Not an easy decision, is it? Fortunately, there are visualization tools that can help you explore these options to make a good choice.

If you agree to participate in this experiment, we will ask you about your experience with data visualizations and you will try out two visualizations to interactively explore a set of options and make your decision. You will receive explanations about the data and visualizations first. To help you make the choice, we will ask you about your preferences. After using the visualizations, you will briefly reflect on the choices you made.

The duration of this study is around 40 minutes.

If possible, the experiment should be conducted using a desktop PC. Please also make sure to maximize the browser window that displays the study.

Voluntariness and Confidentiality

Your involvement in the experiment is voluntary and you are free to withdraw at any time without giving any reason.

Your answers will be treated confidentially and will only be accessible to researchers for scientific evaluation and publication of the results in anonymized form.

If you have questions or concerns please anonymously contact the experimenter via Prolific.

I confirm that I have read and understood the above participant information for the experiment.

I do not want to participate in the above experiment.

I agree to take part in the above experiment.

Survey Navigation

Please only use the survey navigation buttons. Do not use the back button of your browser to return to or edit a previous page. If you use the browser back button, your answers might not be saved.

Experience with Data Visualizations

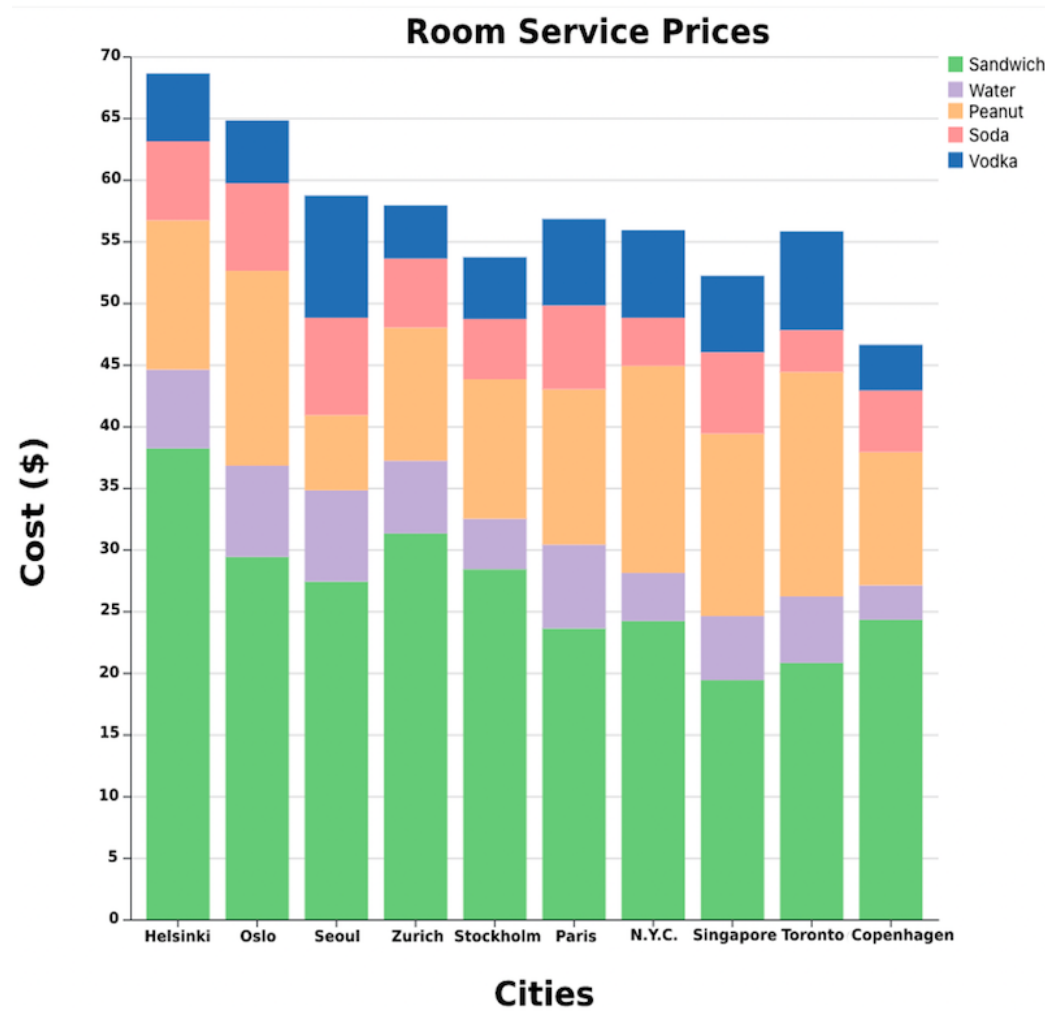
Before we start, we would like to learn about your general background in reading and interpreting different visualizations. This will take not more than 5 minutes.

In the following, you will see 12 static data visualizations. Each visualization comes with a multiple-choice question about the depicted data. Choose the correct answers to the best of your ability. If you are unsure about the correct answer, select "Skip" instead of guessing.

You will have 25 seconds to answer each question. When the time is up, the system will skip the question and automatically proceed to the next one. Don't worry, if you do not get some questions answered in time.

Remaining time: -18:28

1. What is the cost of peanuts in Seoul?



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\$ 5.2

\$ 6.1

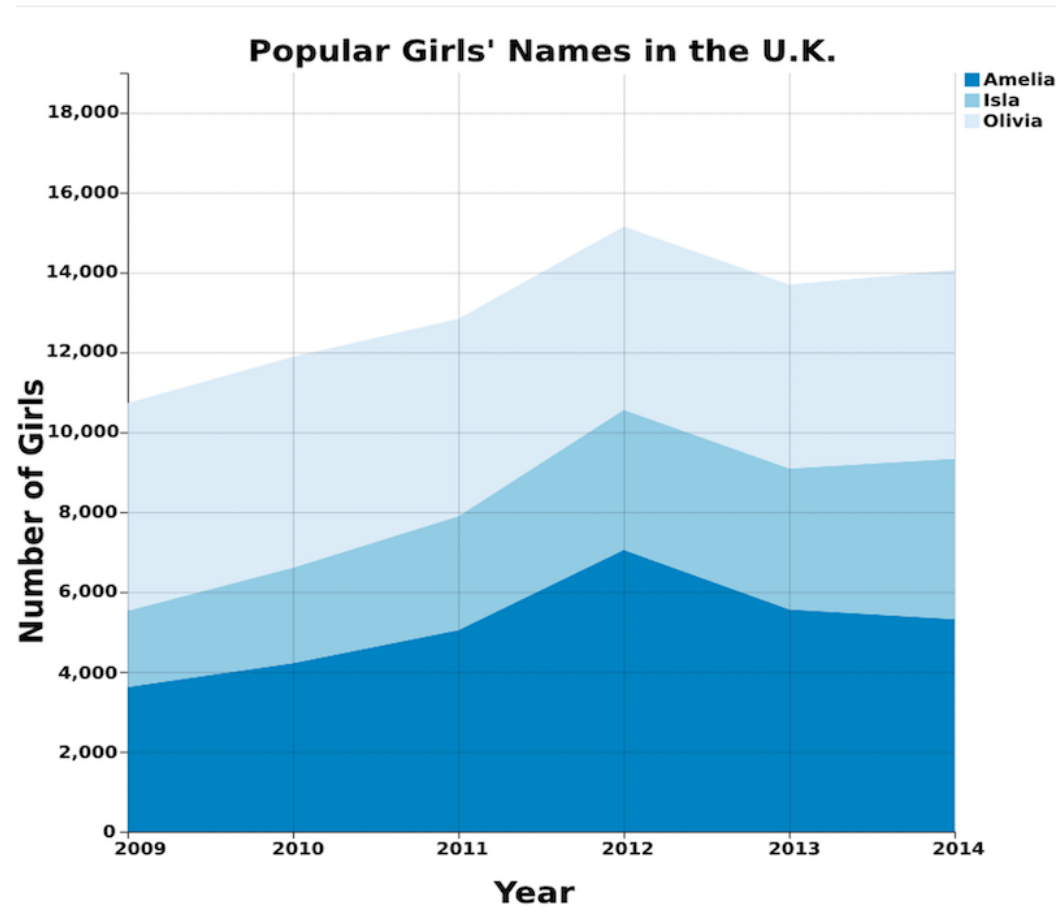
\$ 7.5

\$ 4.5

Skip

Remaining time:

2. What was the ratio of girls named “Isla” to girls named “Amelia” in 2012 in the UK?



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1 to 1

1 to 2

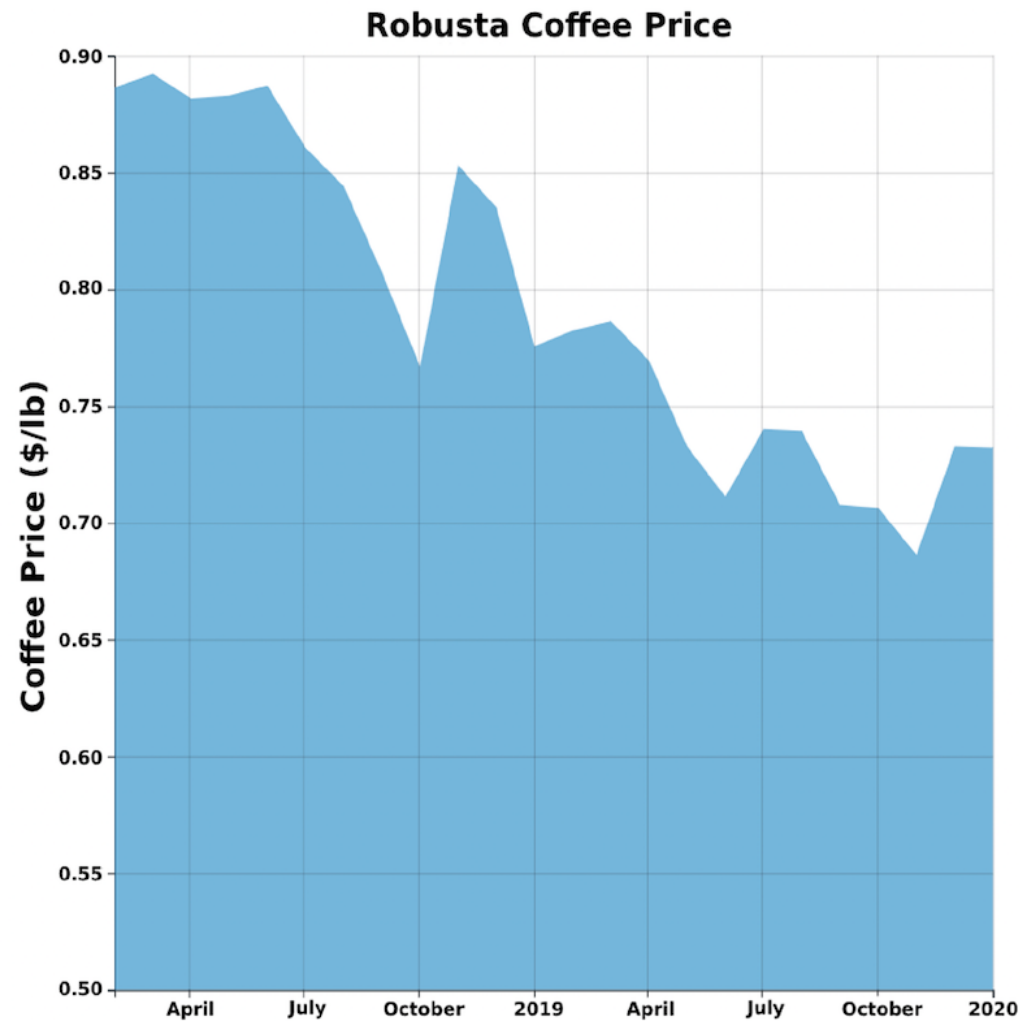
1 to 3

1 to 4

Skip

Remaining time:

3. What was the average price of a pound of coffee in October 2019?



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\$ 0.71

\$ 0.90

\$ 0.80

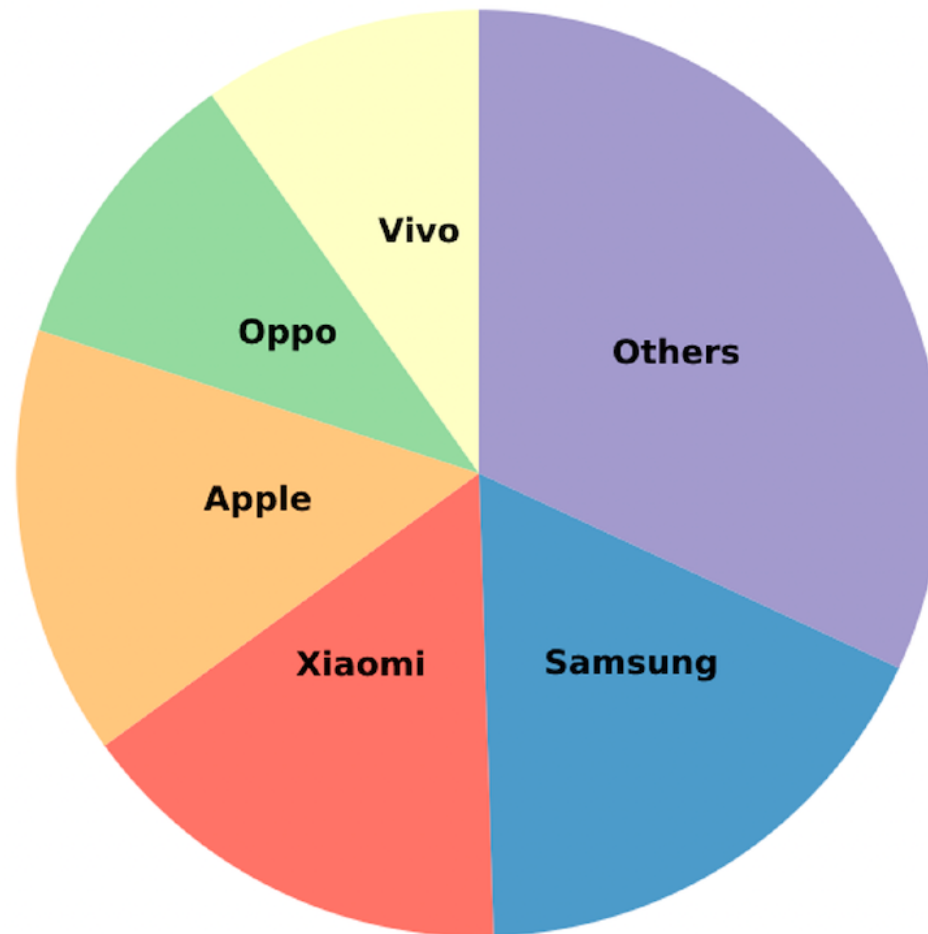
\$ 0.63

Skip

Remaining time:

4. What is the approximate global smartphone market share of Samsung?

Global Smartphone Market Share in 2021



[Visualization](#) by Saugat Pandey and Alvitta Ottley is licensed under [CC BY-NC](#).

17.6 %

25.3 %

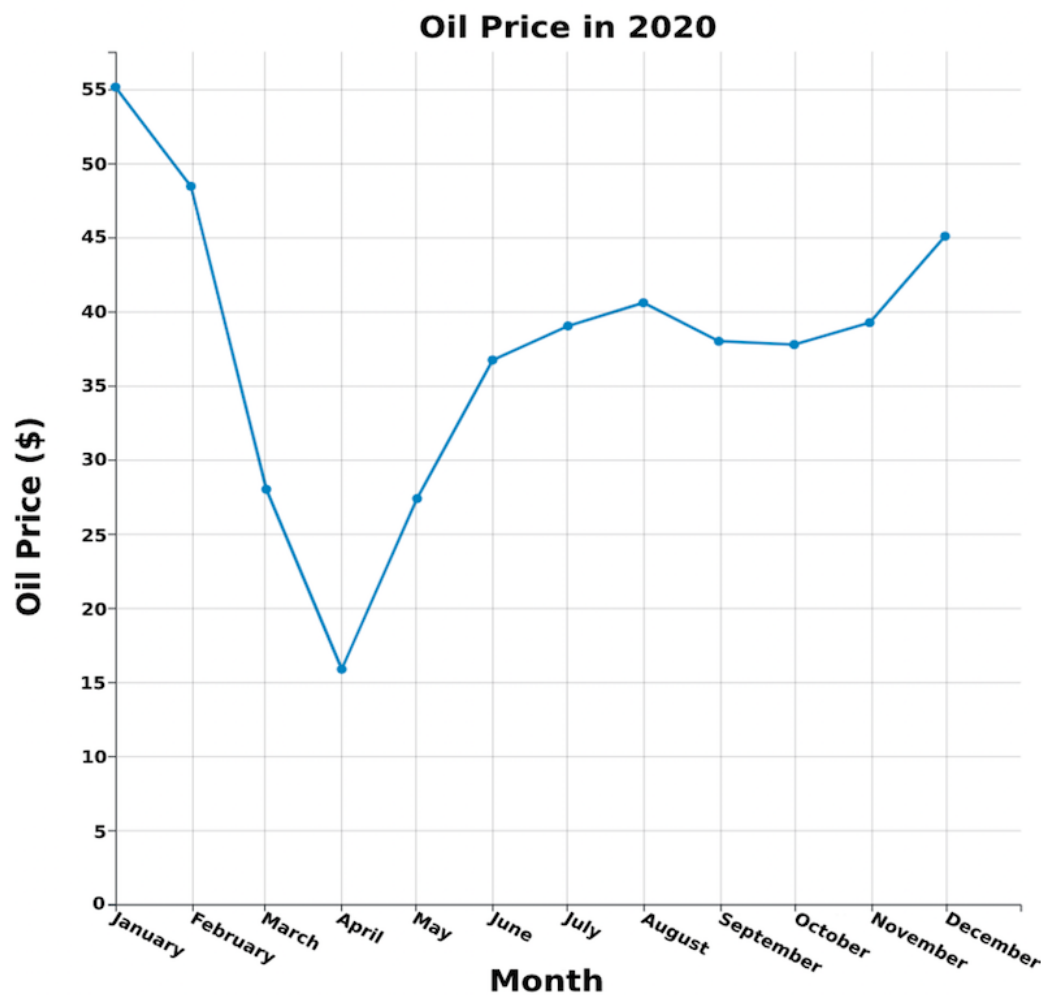
10.9 %

35.2 %

Skip

Remaining time:

5. What was the price of a barrel of oil in February 2020?



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\$ 50.54

\$ 47.02

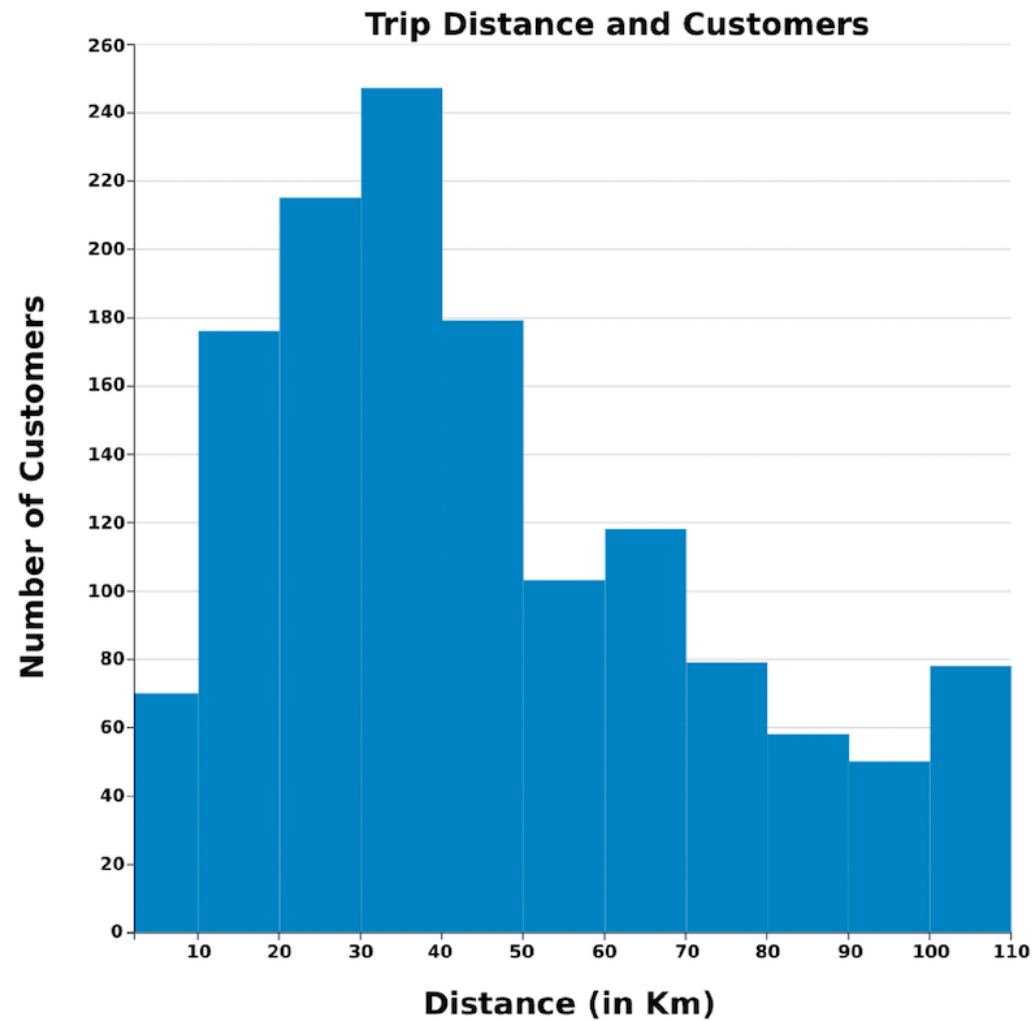
\$ 42.34

\$ 43.48

Skip

Remaining time:

6. What distance have customers traveled in the taxi the most?



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60 – 70 km

30 – 40 km

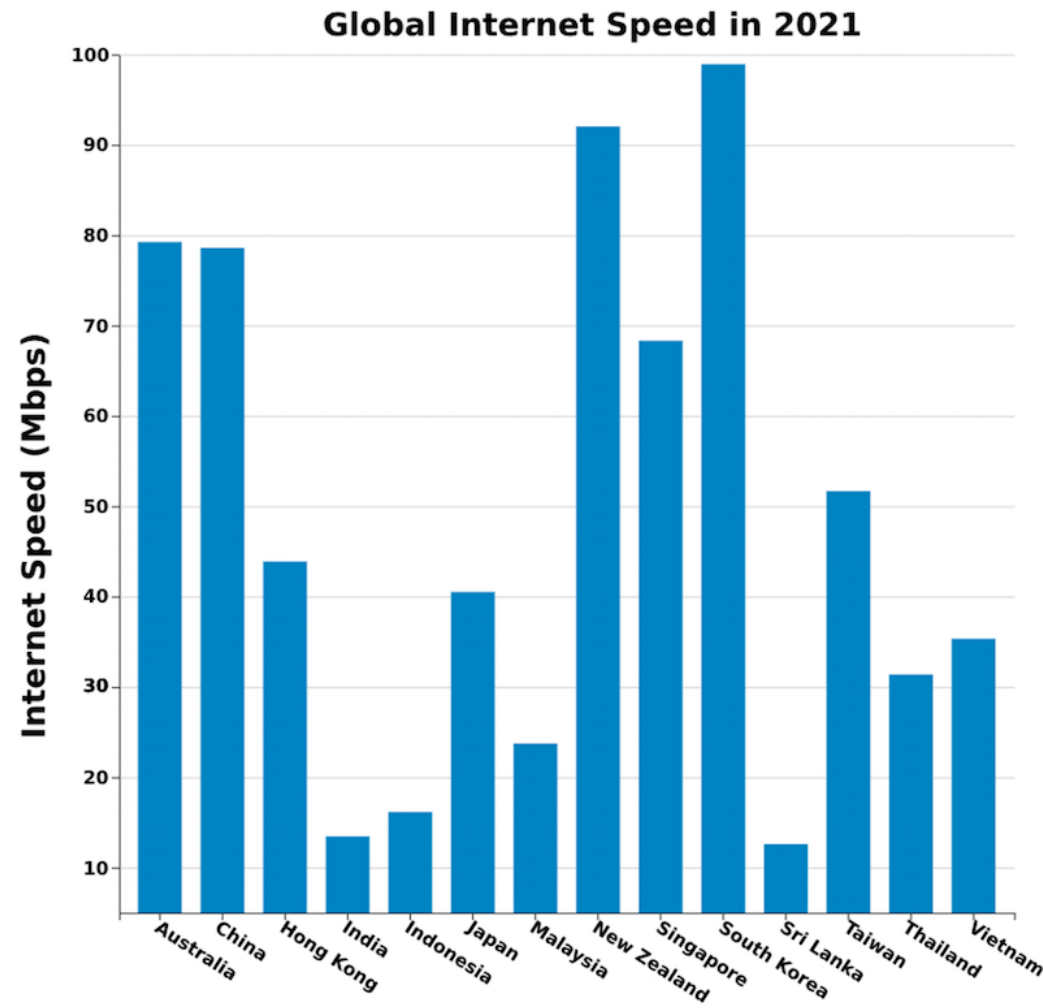
20 – 30 km

50 – 60 km

Skip

Remaining time:

7. What is the average internet speed in Japan?



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42.30 Mbps

40.51 Mbps

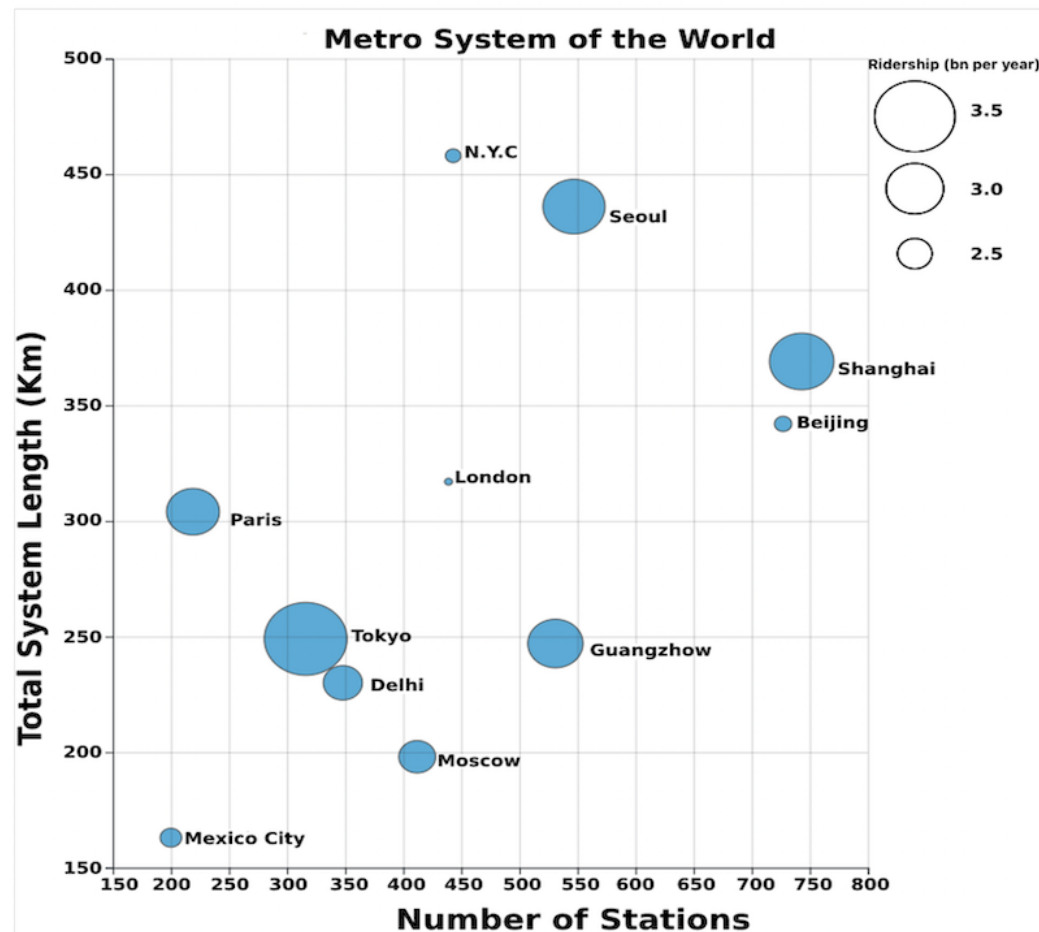
35.25 Mbps

16.16 Mbps

Skip

Remaining time:

8. Which city's metro system has the largest number of stations?



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Beijing

Shanghai

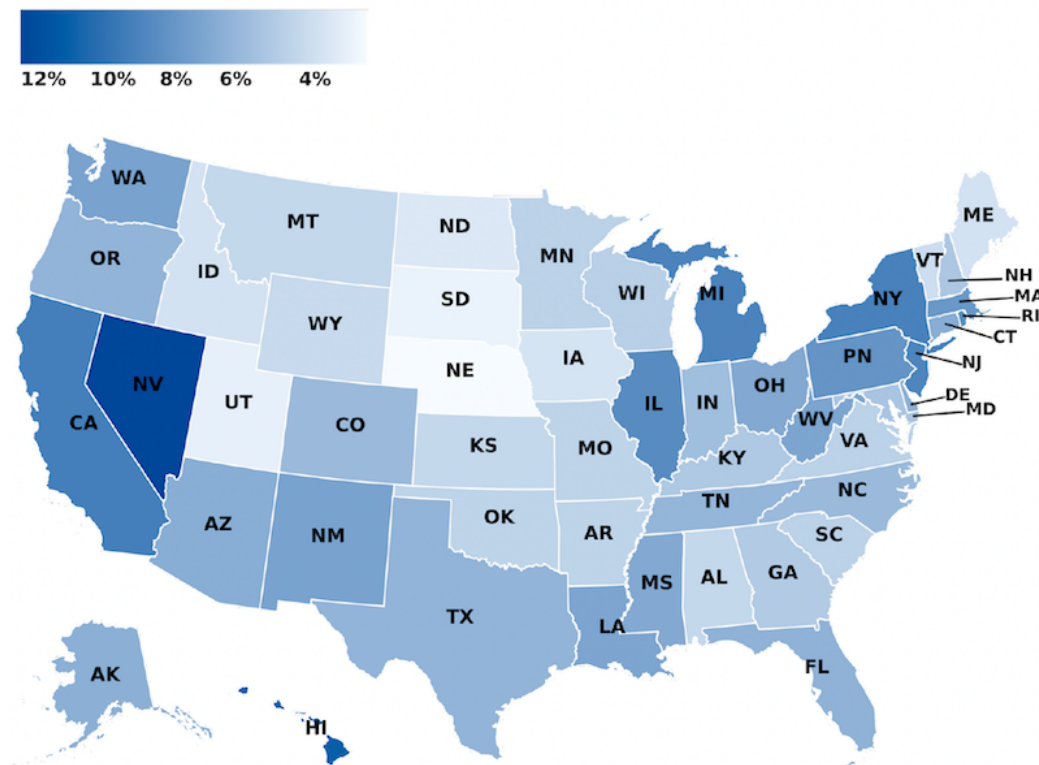
London

Seoul

Skip

Remaining time:

9. In 2020, the unemployment rate for Washington (WA) was higher than that of Wisconsin (WI)?



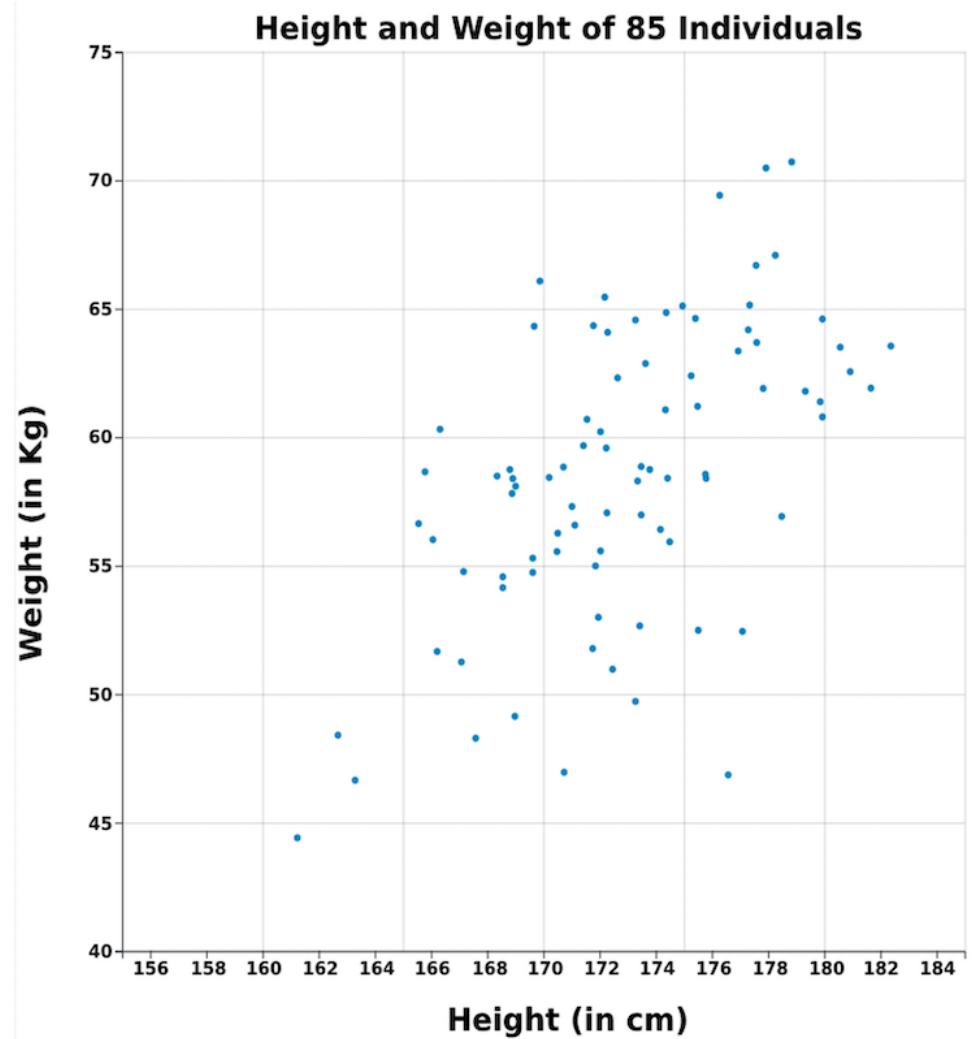
True

False

Skip

Remaining time:

10. There is a negative relationship between the height and the weight of the 85 males.



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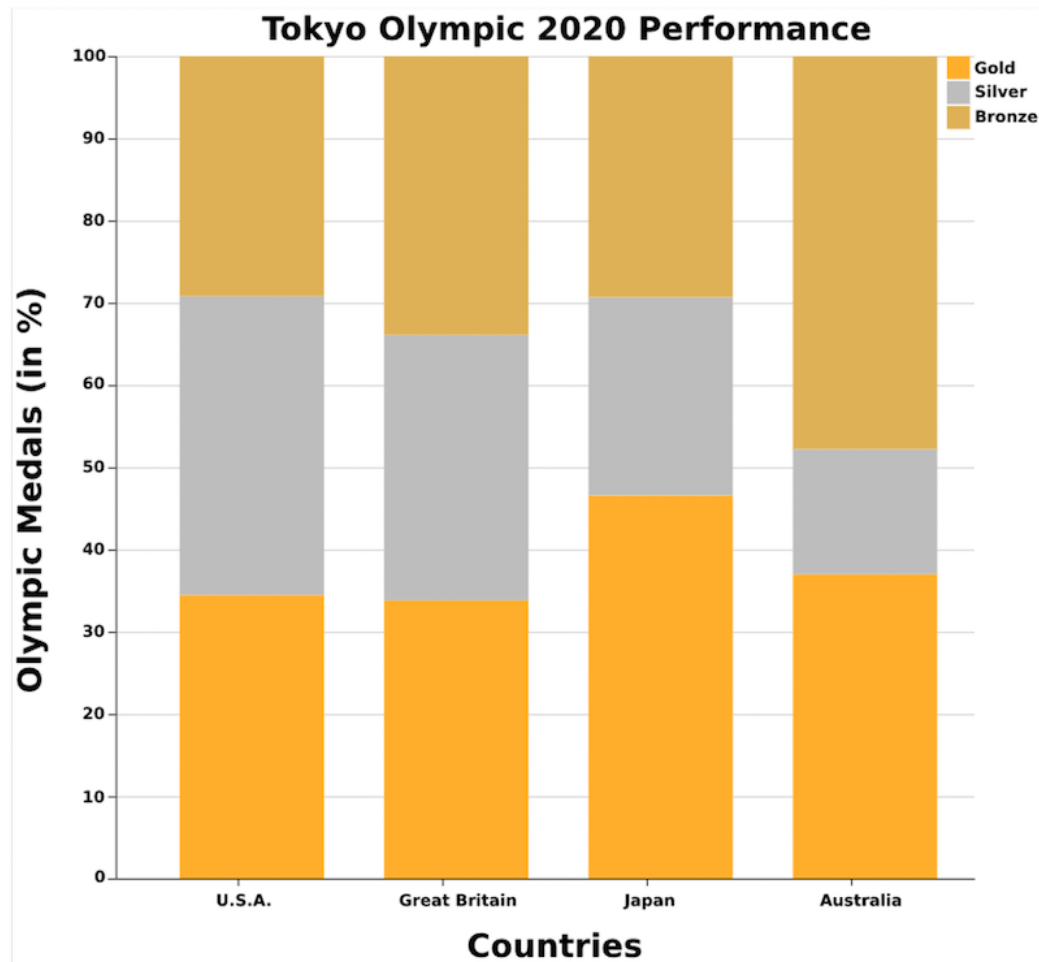
True

False

Skip

Remaining time:

11. Which country has the lowest proportion of Gold medals?



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Great Britain

USA

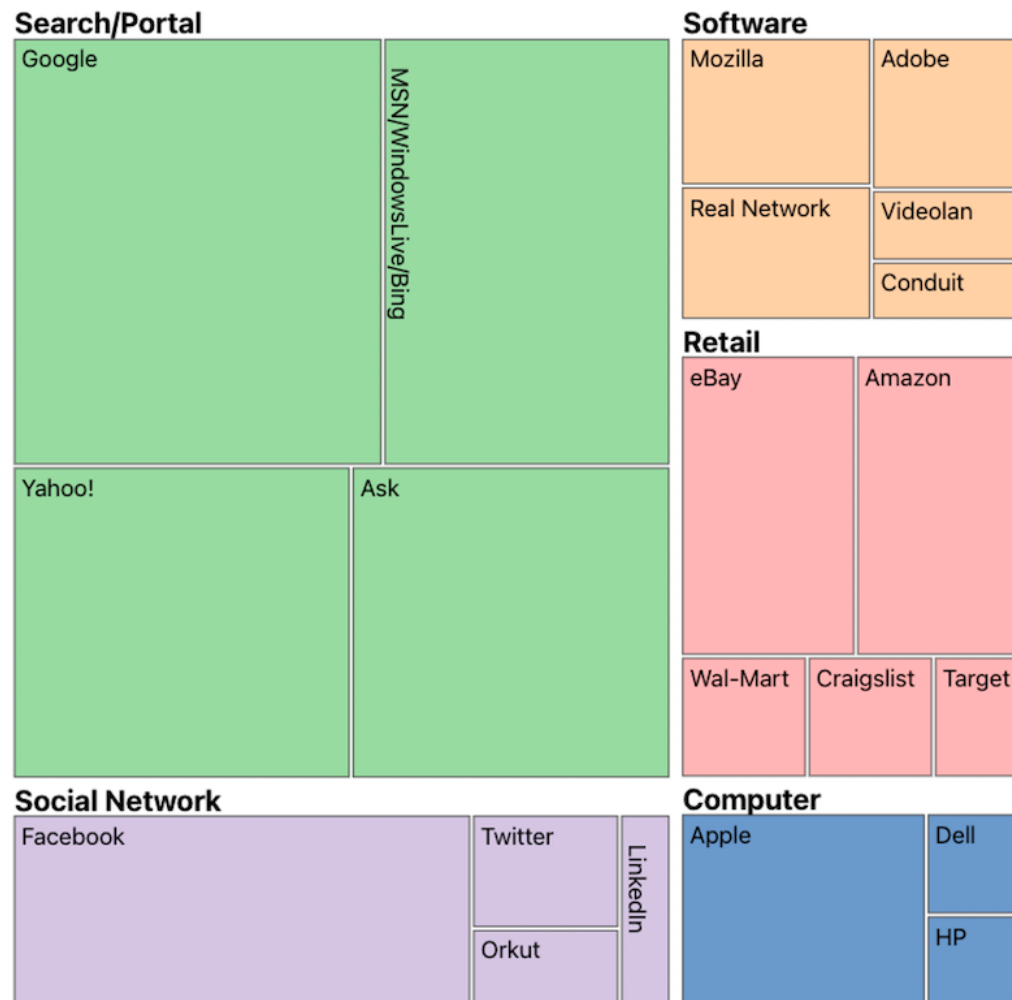
Japan

Australia

Skip

Remaining time:

12. eBay is nested in the Software category.



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True

False

Skip

Experience with Data Visualizations: Success!

You have successfully completed the questions on your experience with data visualizations!

We will now proceed with the decision-making. We will first guide you through a warm-up so you can familiarize with the user interfaces on an exemplary dataset. Afterwards, you will perform the actual decision tasks on a different dataset.

Warm-Up

You are looking for the nicest hotel for your holiday. For each hotel, you know eight properties: price, quality, archaeology, landscape, night life, security, sport activity, and child friendliness. Prices are per person and day, the currency can be ignored for now. All other properties are ratings from 0 to 100.

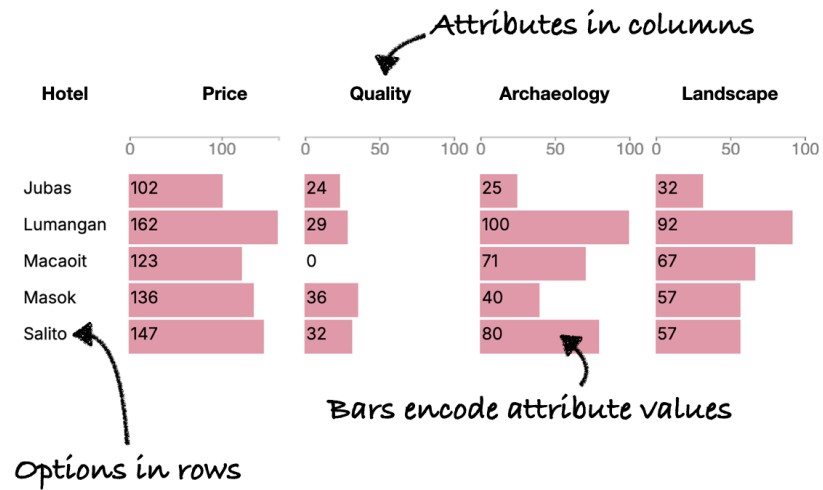
There are two visualizations to help you make such a decision: a tabular visualization and parallel coordinates.

Branching Start

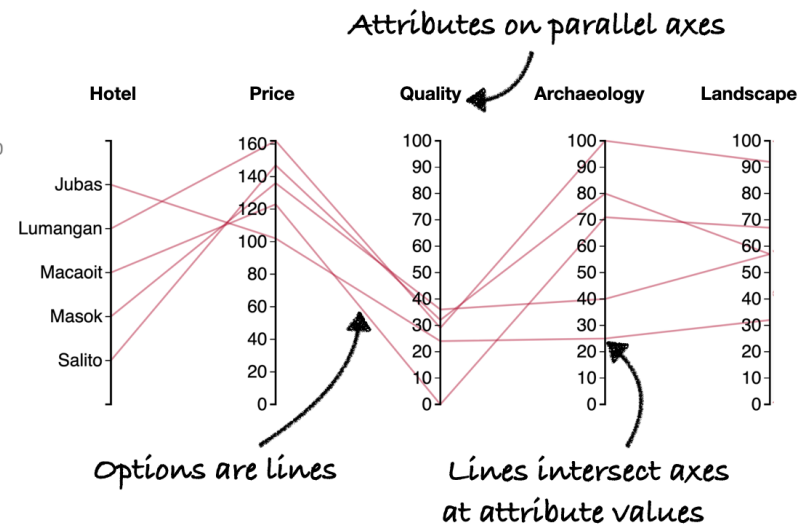
Low Expressiveness Condition

Hotel	Price	Quality	Archaeology	Landscape
Jubas	102	24	25	32
Lumangan	162	29	100	92
Macaoit	123	0	71	67
Masok	136	36	40	57
Salito	147	32	80	57

Tabular visualization



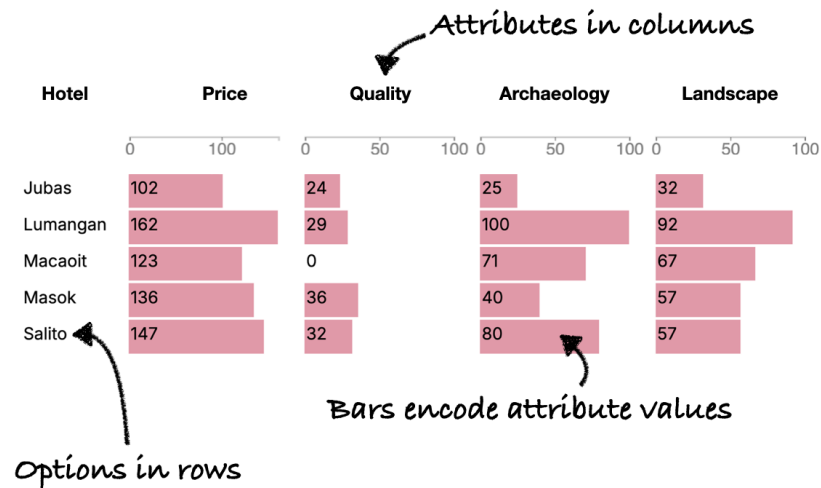
Parallel coordinates



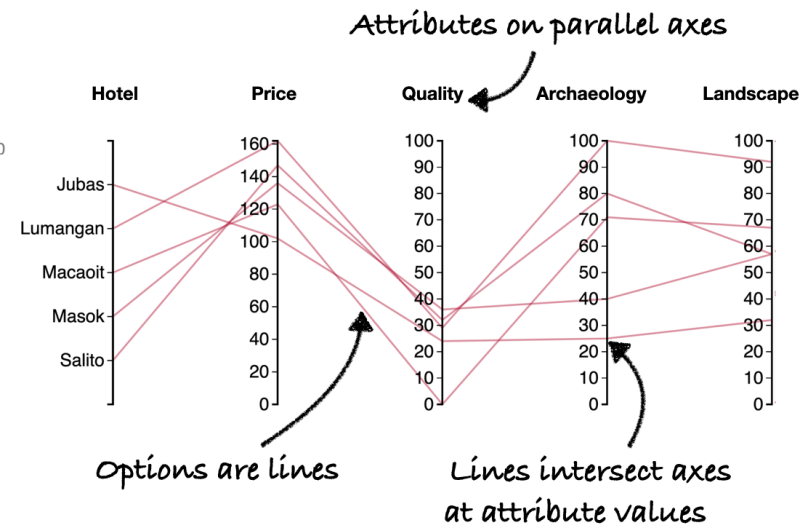
Medium Expressiveness Condition

Hotel	Price	Quality	Archaeology	Landscape
Jubas	102	24	25	32
Lumangan	162	29	100	92
Macaoit	123	0	71	67
Masok	136	36	40	57
Salito	147	32	80	57

Tabular visualization



Parallel coordinates



To effectively help you decide, however, it is important that you tell us what you prefer or aim to avoid when searching for a hotel. Prior to a decision, we will therefore ask you to express your preferences by defining a function that maps property values to preference scores.

Let us have a look at the function editor, before we proceed to the visualization training.

Warm-Up: Preference Function Editor

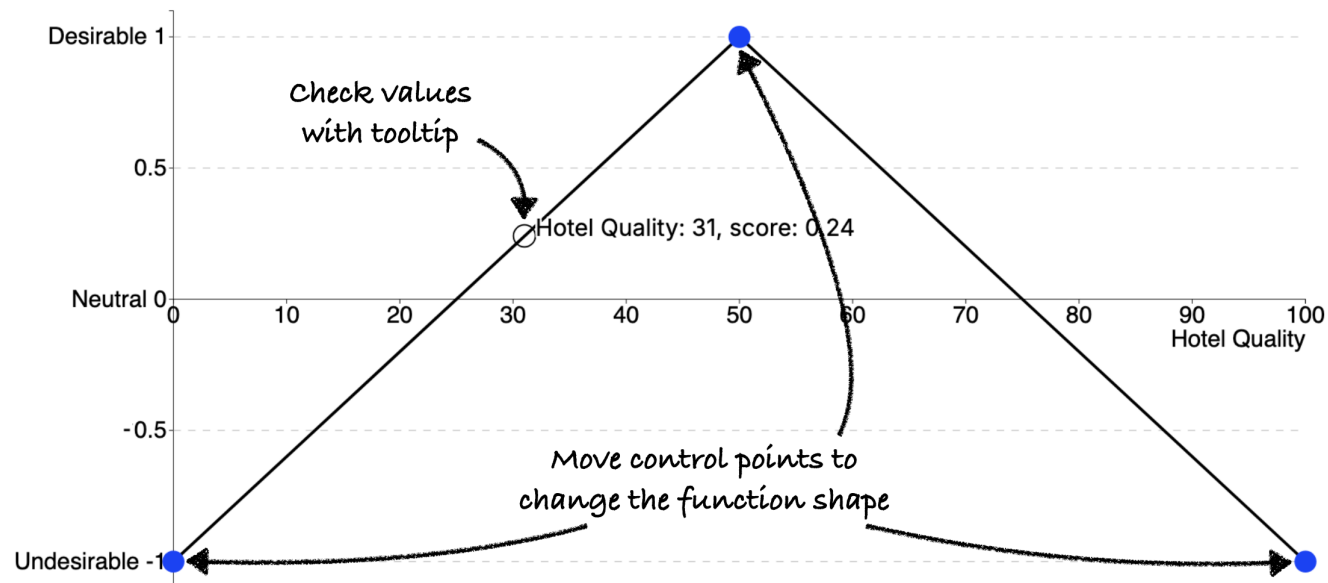
By interactively modifying the shape of the function, you can specify which property values you prefer and how much.

On this page, we cover the basic functionality of the function editor and how to interpret the function shape. On the next page, you will get to know the different interactions that allow you to modify the shape.

Explanation

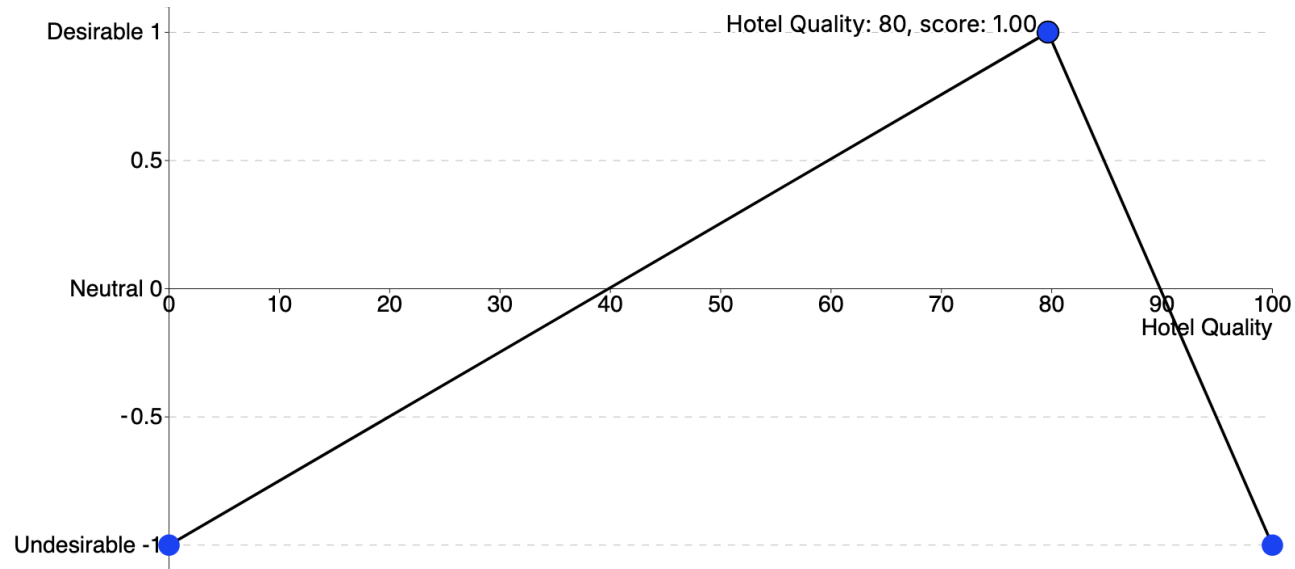
Mirror function at horizontal axis
 ↩ Flip

Remove curve and start from scratch
 ↻ Start over



Example

Alice would like a hotel with a quality ranking of around 80 best. While it should not be much larger, a little lower would also be OK for her to save some money. A hotel with a quality ranking of less than 40 or more than 90 is increasingly undesirable.



Up Next: Interaction Tutorial

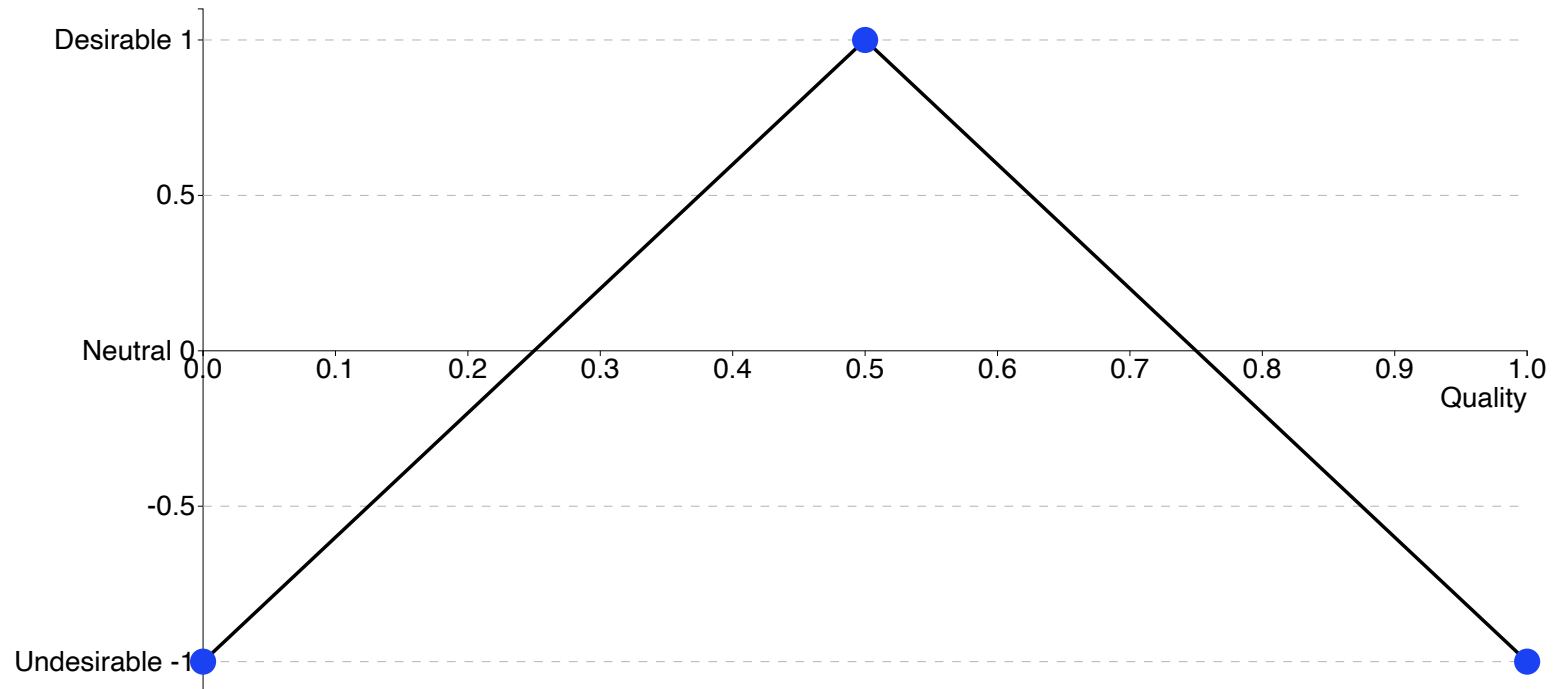
In the following, you can try out the editor yourself.

Warm-Up: Interaction with Preference Function Editor

You can **drag** the points to move them. The middle point can be moved freely, the start and end points can only be moved vertically.

Use the buttons to **flip** the current line vertically or to reset the line to its initial state so you can **start over**.

↕ Flip ↻ Start over



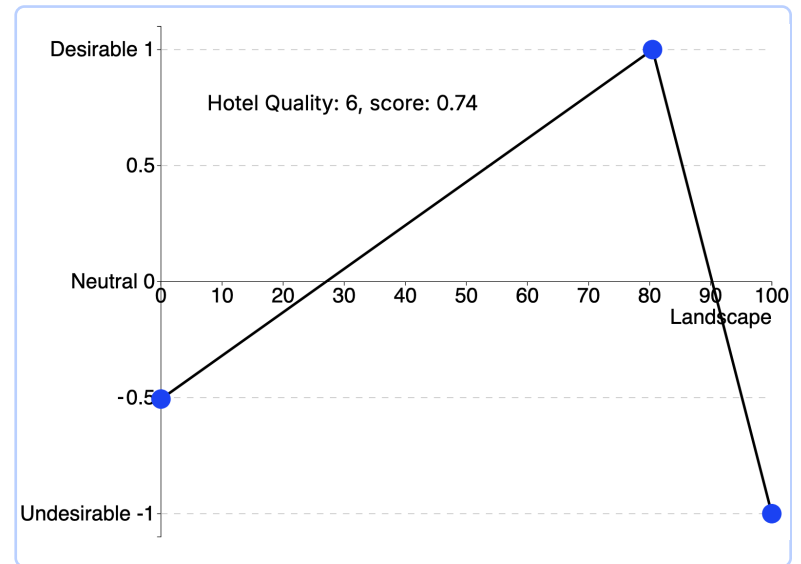
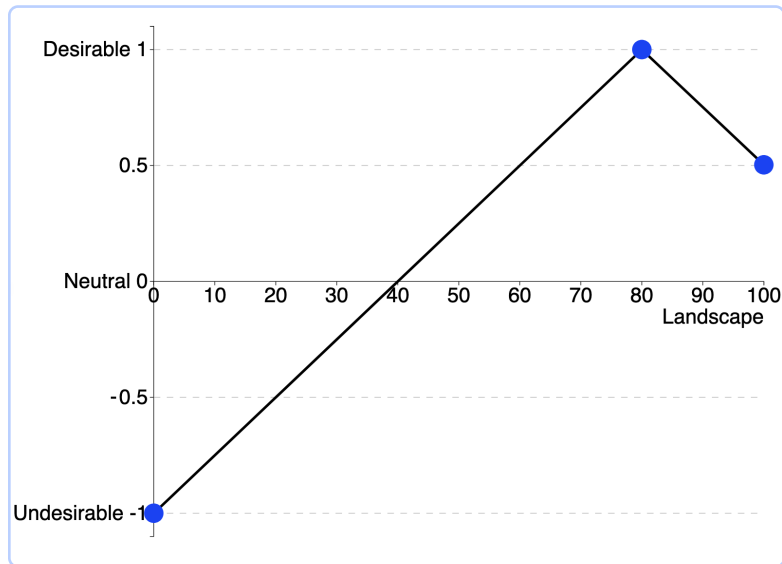
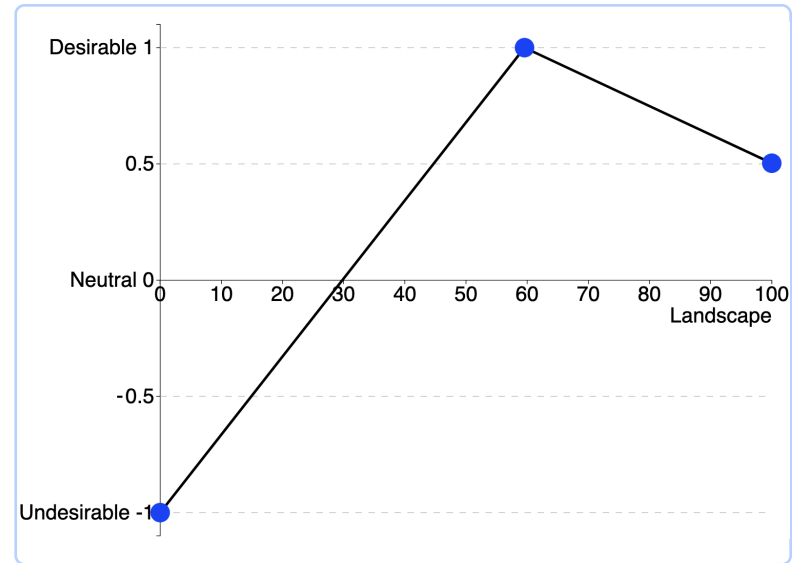
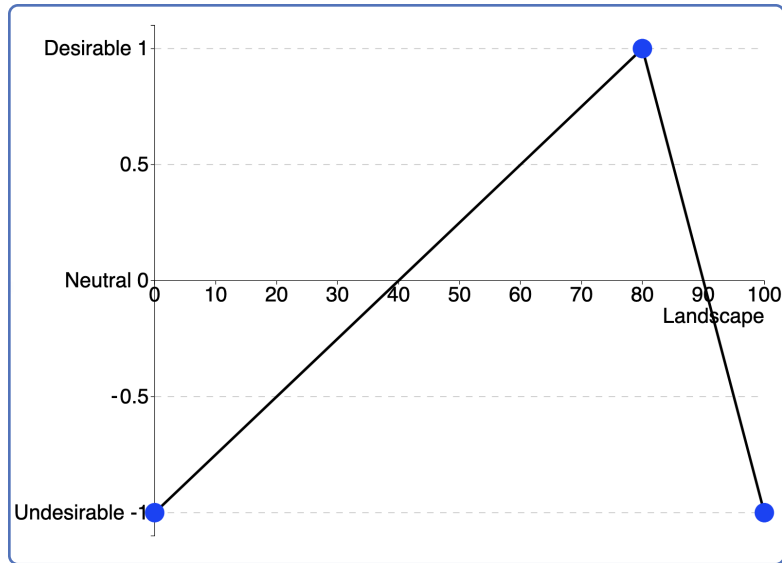
Up Next: Sample Task

In the following, we have prepared a sample task for you.

Warm-Up: Preference Function Editor - Sample Task

Alice would like a hotel that has a solid landscape ranking to enjoy outdoor activities during her vacation. Ideally, the hotel should have a ranking of about 80. A higher ranking is also rather preferred, but she worries that this might come with drawbacks like a difficult arrival or isolation. Lower landscape rankings are increasingly less preferred, with a ranking of 40 or less being undesired.

13. Which of the following functions best reflects Alice's description?



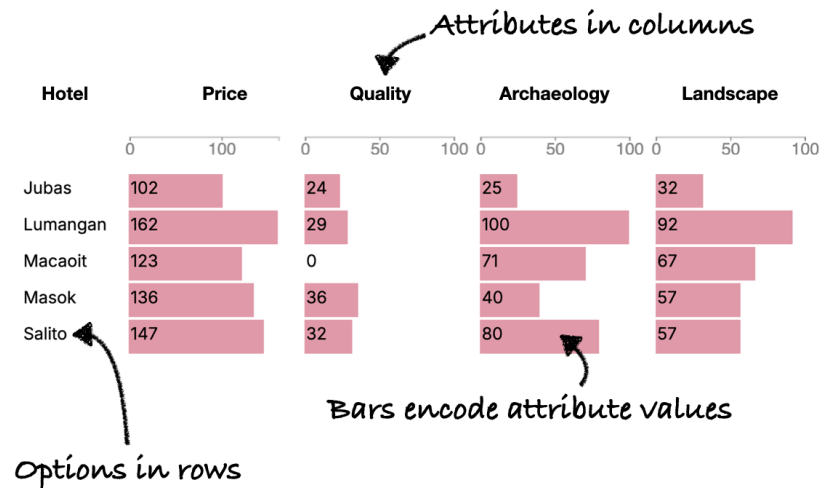
Up Next: Visualization Training

In the following, you will learn how to interact with both visualizations. Let's start with the tabular visualization.

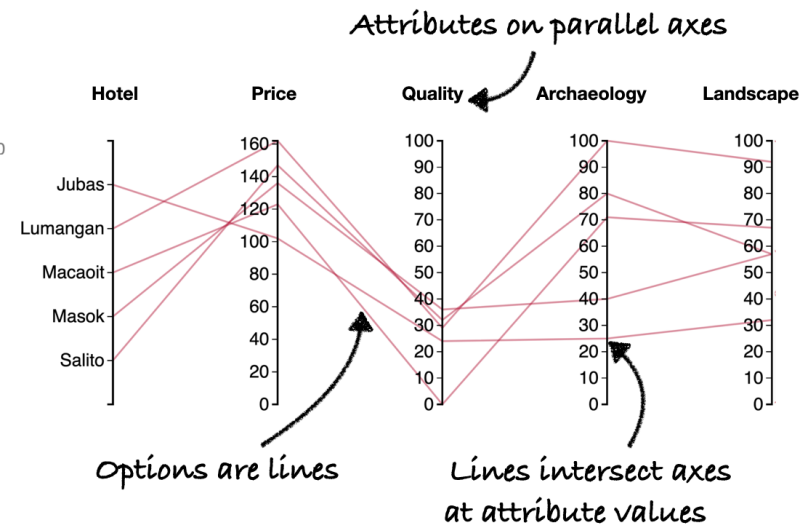
High Expressiveness Condition

Hotel	Price	Quality	Archaeology	Landscape
Jubas	102	24	25	32
Lumangan	162	29	100	92
Macaoit	123	0	71	67
Masok	136	36	40	57
Salito	147	32	80	57

Tabular visualization



Parallel coordinates



To effectively help you decide, however, it is important that you tell us what you prefer or aim to avoid when searching for a hotel. Prior to a decision, we will therefore ask you to express your preferences by defining a function that maps property values to preference scores.

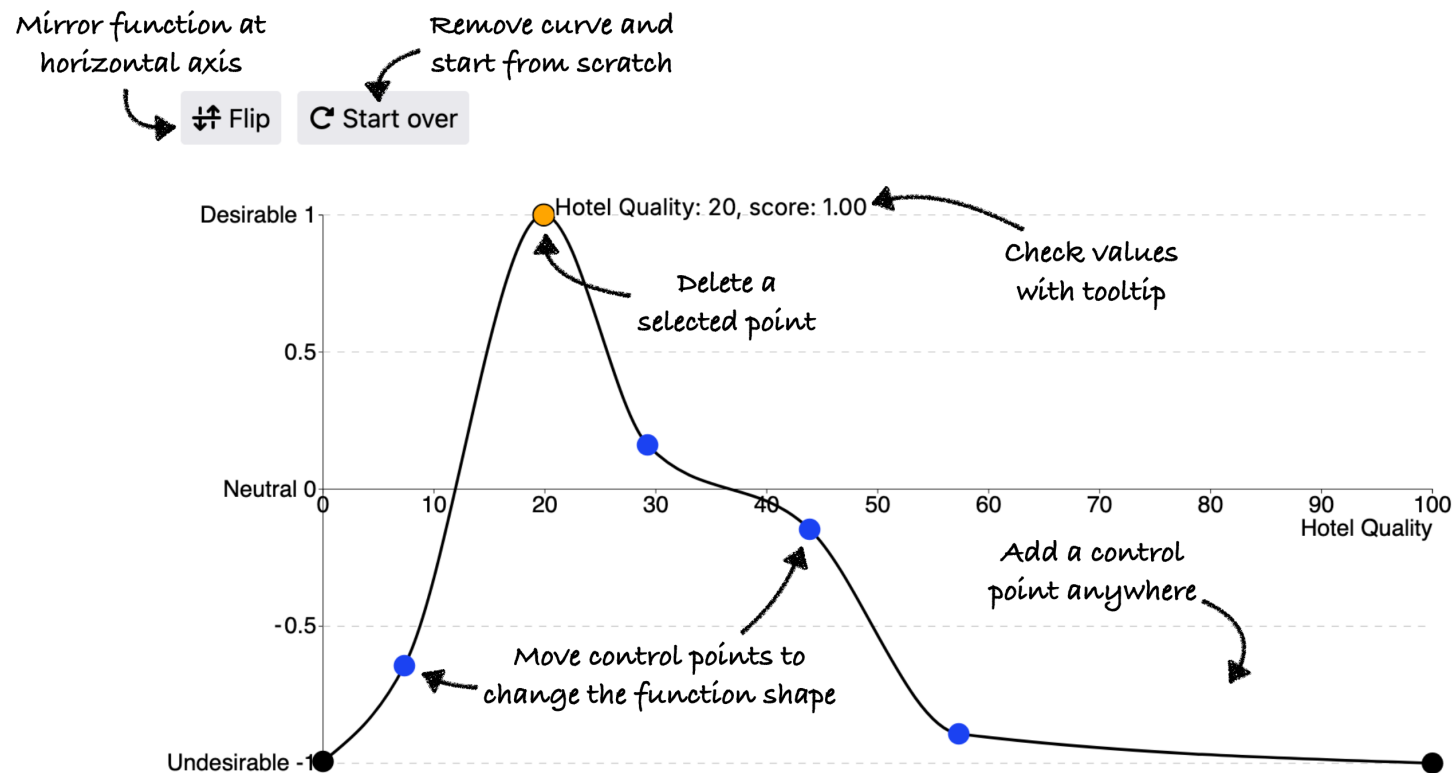
Let us have a look at the function editor, before we proceed to the visualization training.

Warm-Up: Preference Function Editor

By controlling the shape of the function, you can specify which property values you prefer and how much.

On this page, we briefly introduce what you can do with the function editor and how to interpret the function shape. How exactly you can modify the shape is explained on the next page.

Explanation

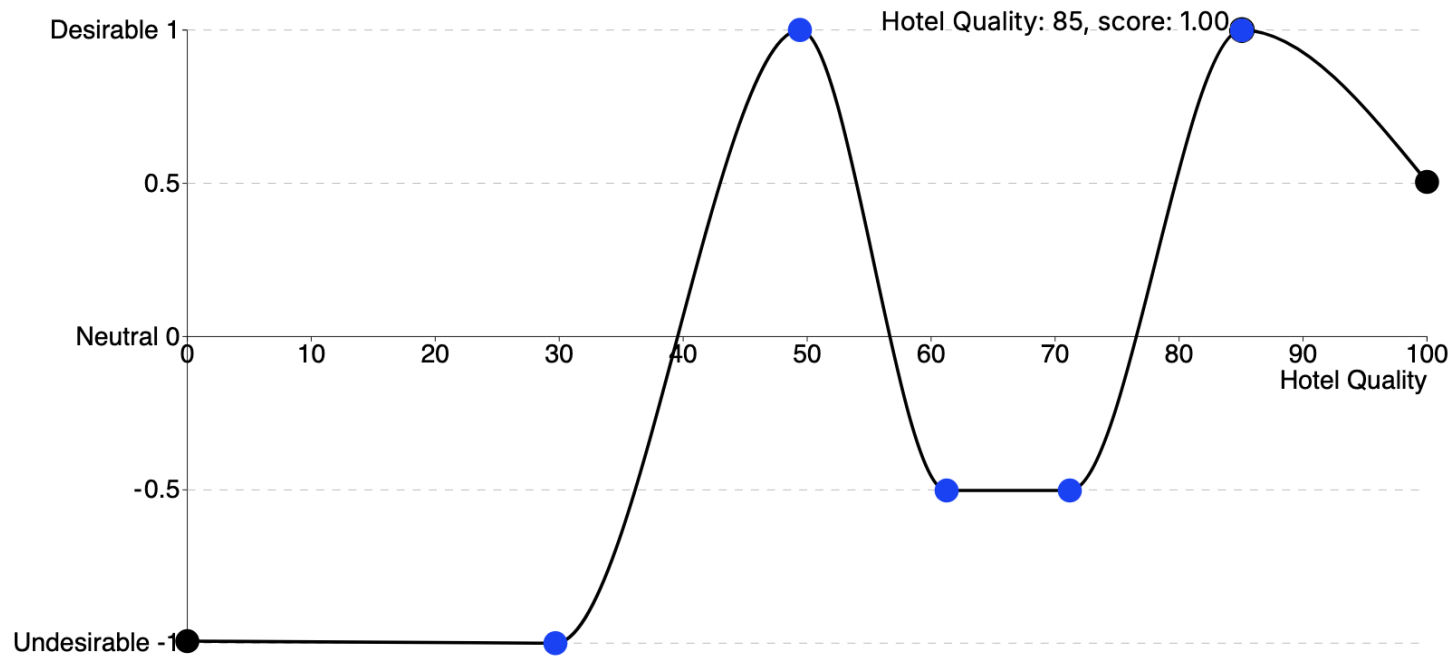


Example

Alice is searching for a hotel to spend her vacation. Her wishes for the hotel's quality rating are as follows:

- Alice would like a hotel with either a moderate quality of around 50 (in that case she would save some money) or a very good quality of around 85.
- An exceptional quality higher than 85 would be OK, but not fully desirable, because she assumes that exceptional hotels might be overpriced.
- A hotel in between is rather undesirable for her, because she assumes that she will not get an appropriate value for money.
- Everything close to and below 30 suddenly becomes unacceptable for her.

Her function would look something like this:



Up Next: Interaction Tutorial

In the following, you can try out the editor yourself.

Warm-Up: Interaction with Preference Function Editor

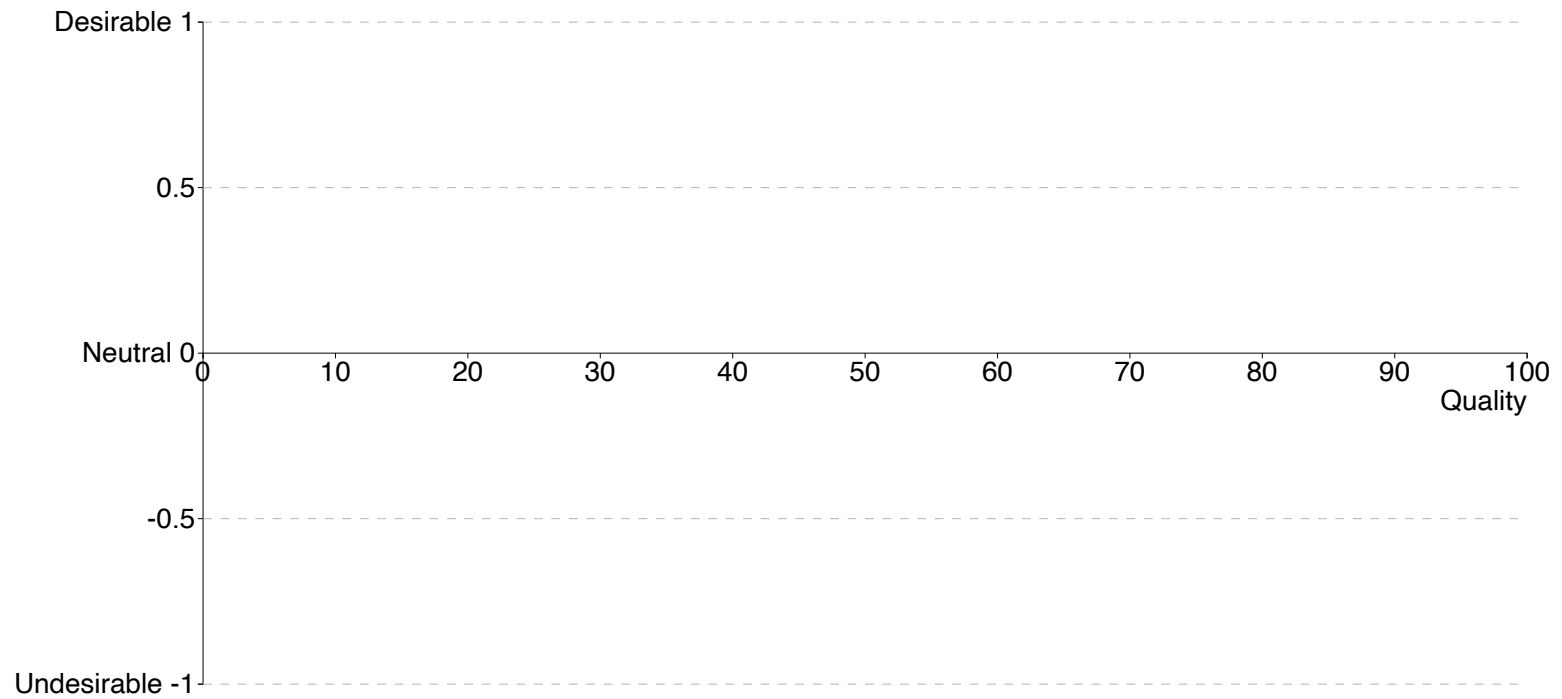
Click anywhere to add points to the curve. The first click automatically completes the line with an additional start point and end point.

You can **drag** any point to move it. The middle points can be moved freely, the start and end points can only be moved vertically.

To remove, **click a point** to select it first (which highlights it in orange) and then use **Backspace** to remove the selected point.

Use the buttons to **flip** the current line vertically or to remove all points so you can **start over**.

↕ Flip ↻ Start over



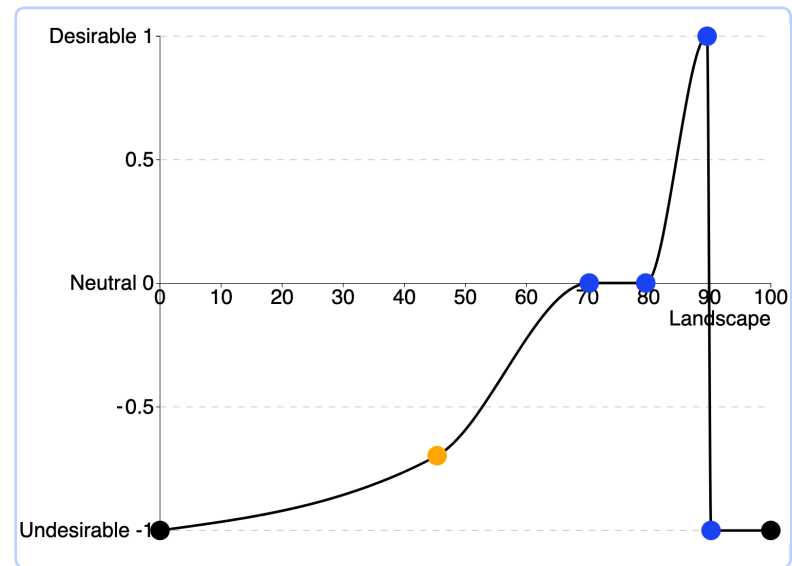
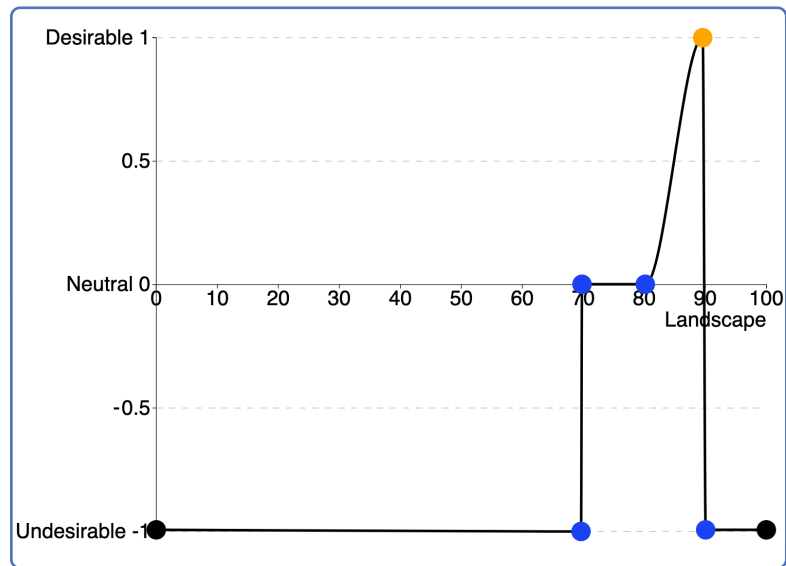
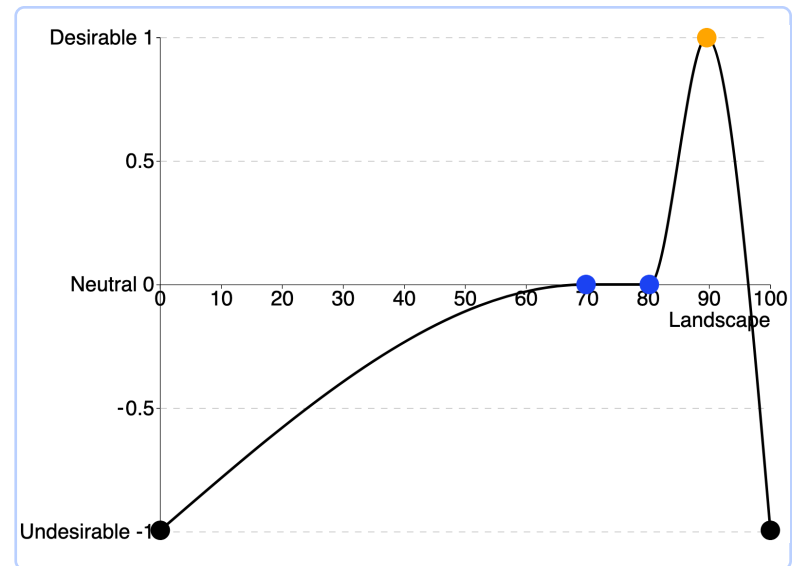
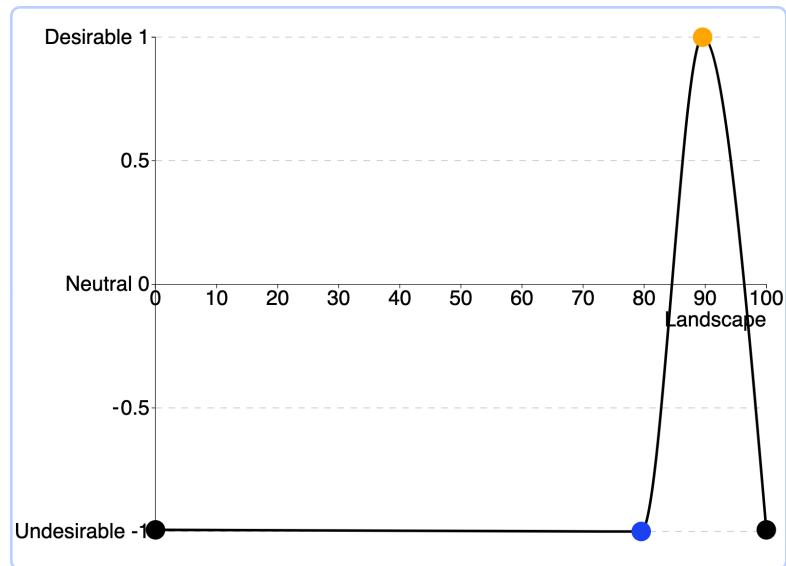
Up Next: Sample Task

In the following, we have prepared a sample task for you.

Warm-Up: Preference Function Editor - Sample Task

Alice would like a hotel that has beautiful landscapes to offer but she also worries about the drawbacks (difficult arrival, isolation, no shopping facilities, etc.) that an accommodation surrounded by untouched nature might cause. Therefore she does not want her hotel to be among those with a landscape ranking larger than 90. At the same time, she prefers to start hiking right from the hotel, the more beautiful the surrounding, the better. Still, everything above a ranking of 70 would be acceptable to her. Rankings below that are strictly to be avoided.

13. Which of the following functions best reflects Alice's description?



Up Next: Visualization Training

In the following, you will learn how to interact with both visualizations. Let's start with the tabular visualization.

Branching End

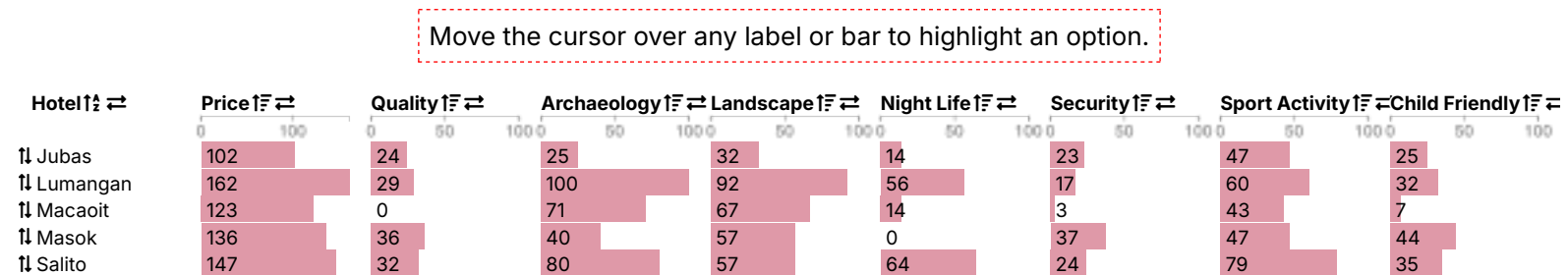
Warm-Up: Tabular Visualization

In the tabular visualization, each option to decide upon is a row. The properties of an option are depicted in the columns. The property values in the cells are visually represented using bars, with the numerical values on top.

1) Highlighting Individual Options

Move the mouse over a bar or label to highlight a single option.

Example

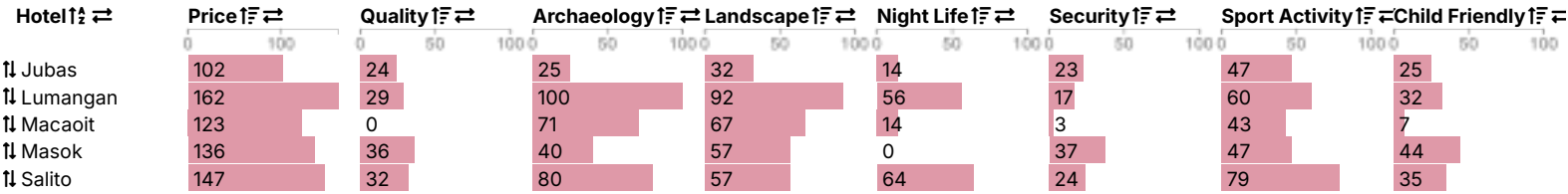


2) Range Selection

Press the mouse button and drag the mouse along an axis to create a range selection that highlights multiple items. The axes are located below the attribute labels. A selection can be resized by dragging one of its ends. It can also be moved along the axis by dragging its center. To remove a selection, simply click anywhere outside the selection on the same axis. When multiple selections exist, each item that fulfills all selected ranges is shown.

Example

Create a range brush on any axis to highlight multiple options.

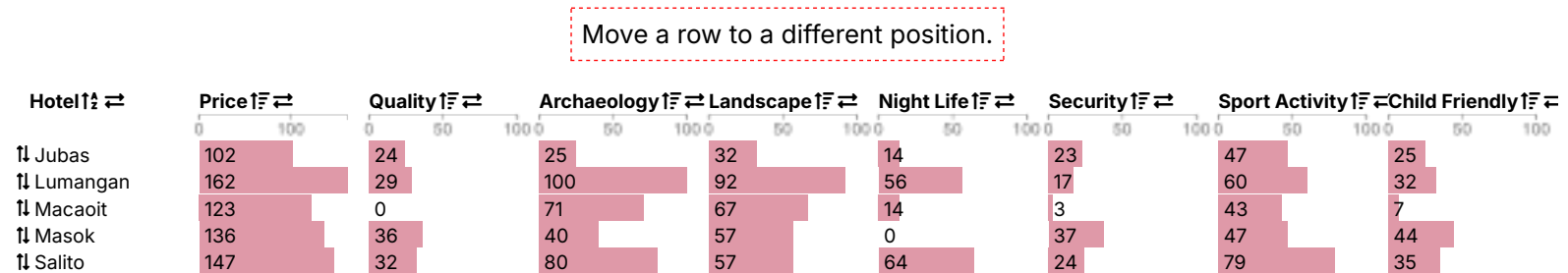


Warm-Up: Tabular Visualization

3) Attribute and Option Reordering

You can drag, move, and drop the arrow icon in a column title  to reorder the columns. You can additionally drag, move, and drop the arrow icon at the beginning of a row  to reorder the rows.

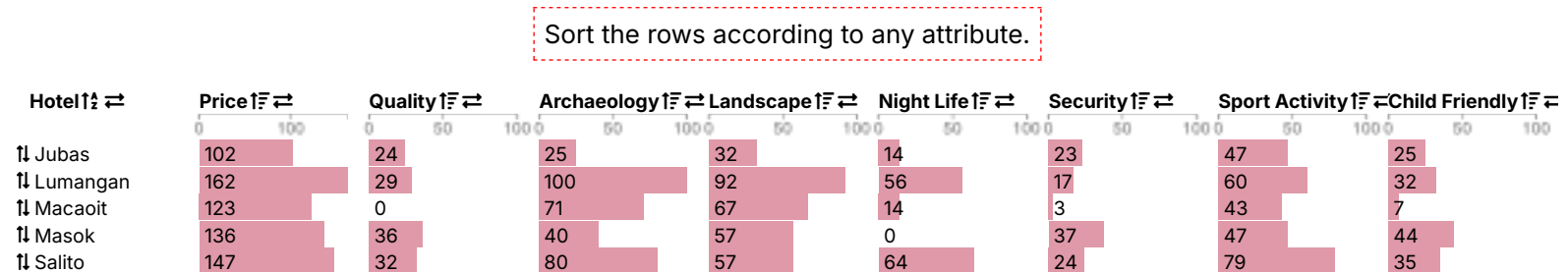
Example



4) Attribute Sorting

You can click the sort icon  in a column title to sort all rows according to this attribute's values.

Example



Up Next: Sample Questions

On the next page, you will try out a larger variant of the tabular visualization yourself. For this, we will show you some sample questions that we would like you to answer with the visualization.

14. What is the name of the hotel with Sport Activity = 54?

How much does it cost?

☒ Sugoy ☐ Soom ☐ Mobobolo

☐ 140 ☐ 105 ☒ 127

15. How many hotels have a Security larger or equal to 70 and a Price lower or equal to 160?

Count

6



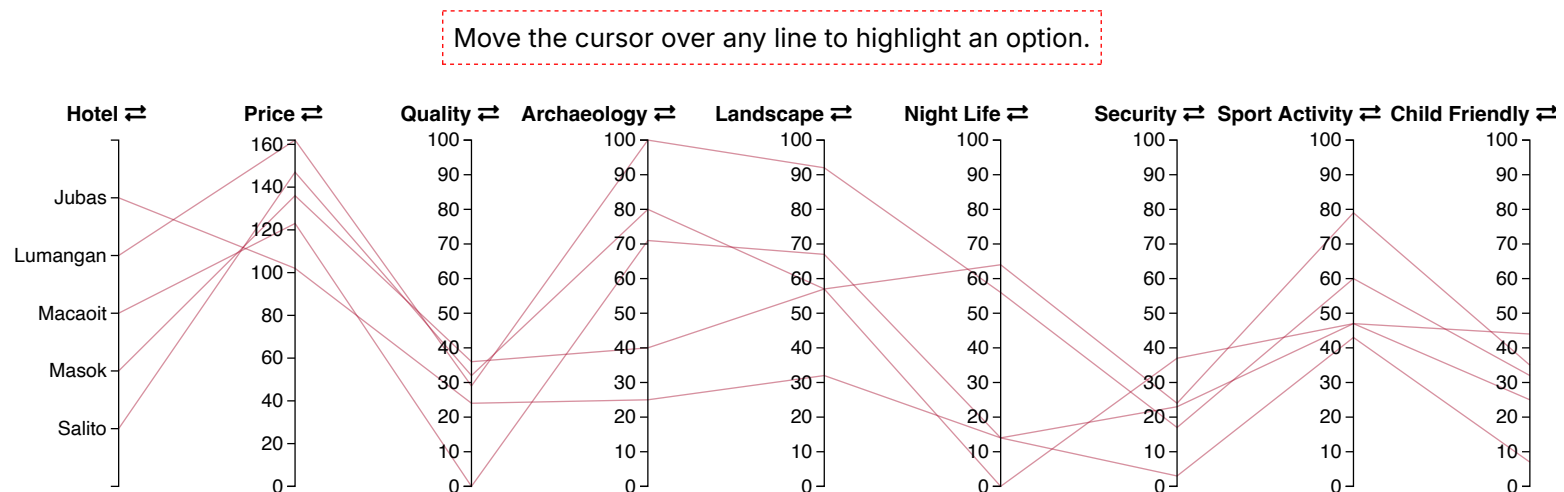
Warm-Up: Parallel Coordinates

In the parallel coordinates, each option to decide upon is a polyline. The properties of an option are depicted on the vertical axes. The polyline intersects the axes at the option's corresponding values.

1) Highlighting Individual Options

You can move the mouse over a line in the parallel coordinates to highlight a single option.

Example



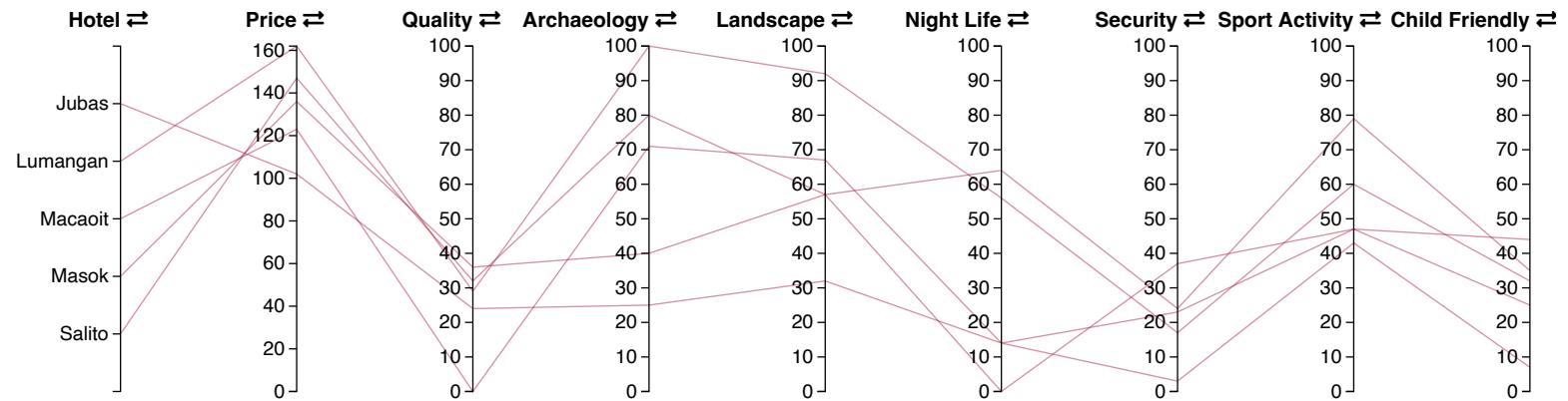
2) Range Selection

The range selection works exactly like before in the tabular visualization, only on the vertical axes this time. Press the mouse button and drag the mouse along any axis (except the first axis with the holiday package names) to create a range selection that highlights multiple options. A selection can be resized by dragging one of its ends. It can also be moved along the axis by dragging its center. To remove a selection, simply click anywhere outside the selection on the same axis. When multiple selections exist, each option that fulfills all selected ranges

is shown.

Example

Create a range brush on any axis to highlight multiple options.

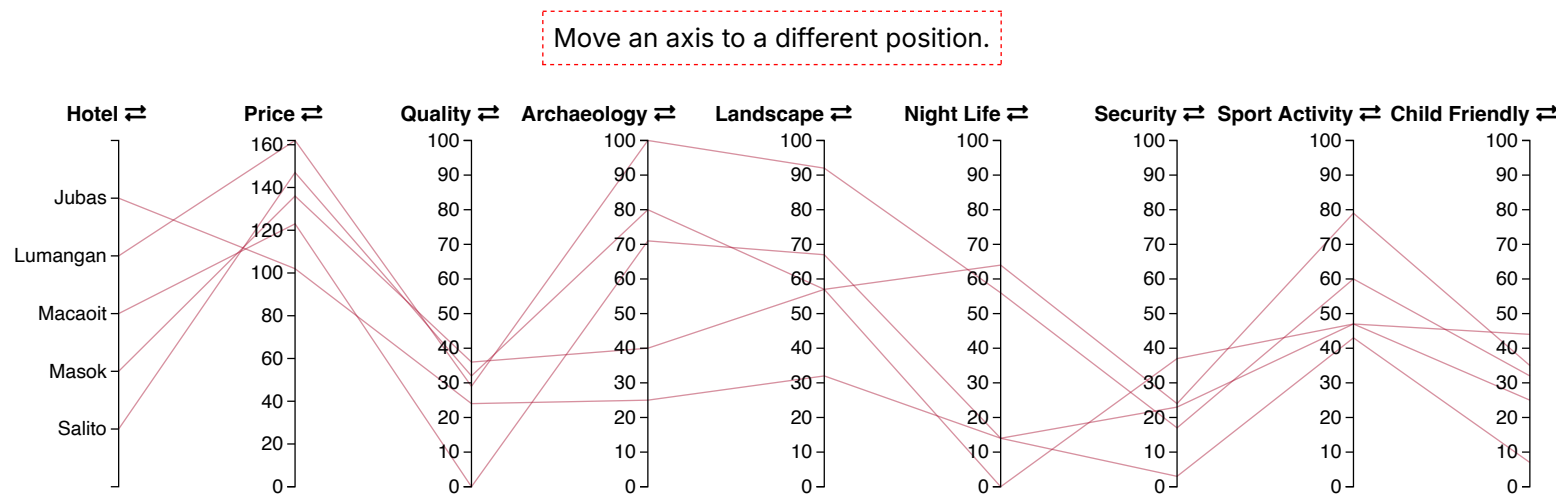


Warm-Up: Parallel Coordinates

3) Attribute Reordering

You can drag, move, and drop the arrow icon ⇌ in an axis title to reorder the axes.

Example



Up Next: Sample Questions

On the next page, you will try out a larger variant of the parallel coordinates visualization yourself. For this, we will show you some sample questions that we would like you to answer with the visualization.

16. What is the name of the hotel with Landscape = 14?

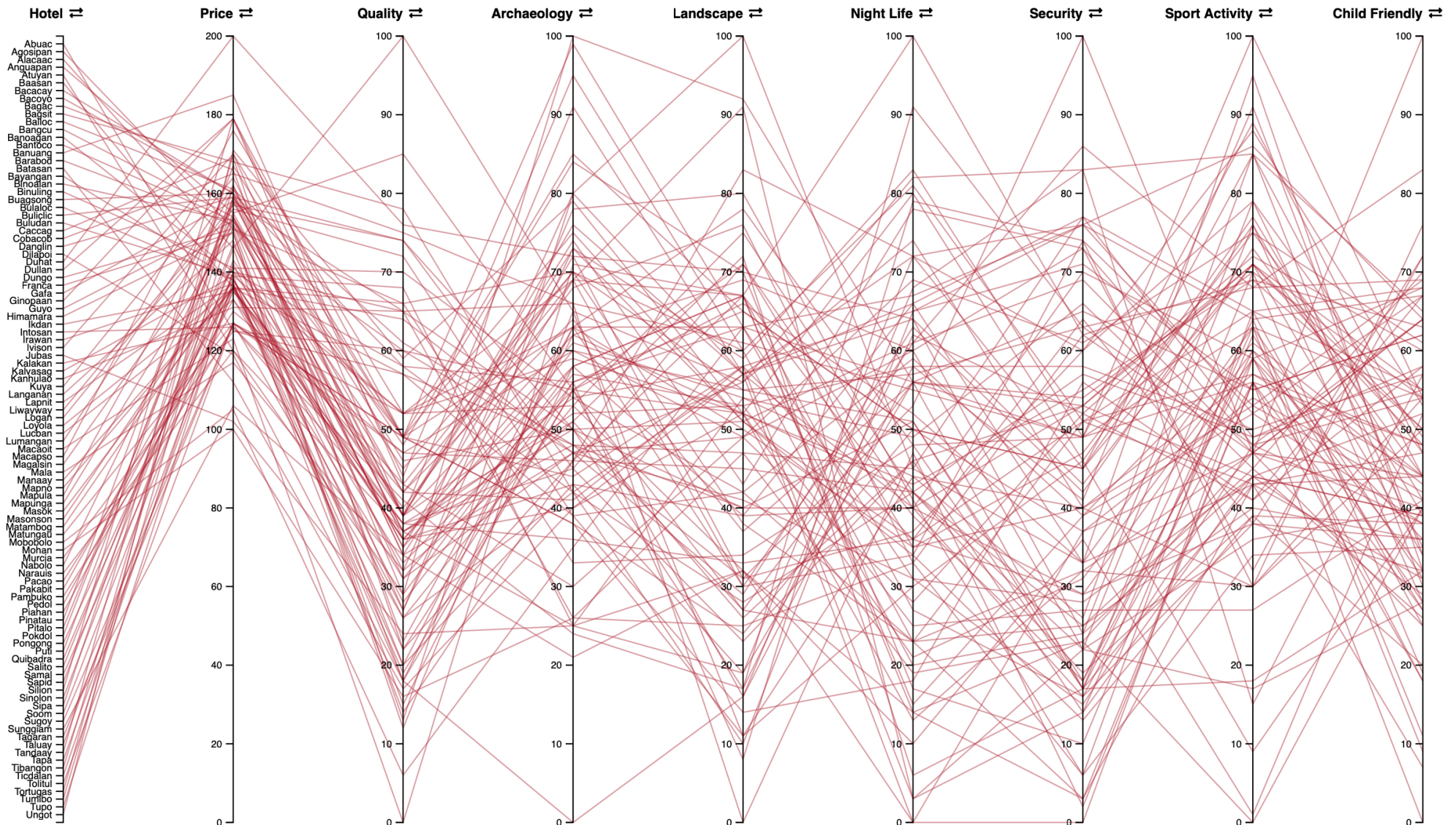
☐ Jubas ☒ Logan ☐ Kalvasag

How much does it cost?

☒ 156 ☐ 102 ☐ 139

17. How many hotels have a Hotel Quality between 70 and 80 and a Child Friendliness between 50 and 60?

Count



Decision Setting

We now proceed to the actual decision. The next questions are all about your own preferences.

Imagine that you are about to move to a different city and you are searching for the perfect apartment to rent. Searching on a real estate platform results in 100 apartments to choose from. For each apartment, it tells you eight properties: year of construction, number of parking spaces, rent, living space, number of rooms, floor, distance to the nearest supermarket, and distance to the nearest public transport station.

Apartment	Year	Park Spaces	Rent (€)	Area (m ²)	Rooms	Floor	Next Supermarket (m)	Next Public Transport (m)
Alisoso	2013	1	1520	144.02	3	3	3454	1318
Banate	1937	1	1225	90	3	2	3519	3005
Salito	1900	1	1792	112	3	1	1491	2977
Managase	2017	2	1100	105.01	3	0	1055	704
Balaorte	1998	1	780	76	3.5	3	3564	388

This scenario uses euro and (square) meters for rent, area, and distance properties, but this is not all-important. If they do not represent units you are familiar with, consider the given spectrum of property values as a general frame of reference.

To help you with the upcoming decision, we need to know your preferences regarding an apartment to rent.

18. When searching for an apartment, how important is to you ...

	Not at all important Extremely important									
... the amount of rent?	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒
... the living area of the apartment?	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐
... the number of rooms?	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐
... which floor the apartment is on?	☐	☐	☐	☐	☐	☒	☐	☐	☐	☐
... the year of construction?	☐	☐	☐	☐	☐	☒	☐	☐	☐	☐
... the number of parking spaces belonging to the apartment?	☐	☐	☐	☐	☐	☐	☒	☐	☐	☐
... the distance to the nearest supermarket?	☐	☐	☐	☐	☐	☐	☐	☒	☐	☐
... the distance to the nearest public transport station?	☐	☐	☐	☐	☐	☐	☐	☒	☐	☐

Branching Start

Low Expressiveness Condition

Decision: Your Preferences

For the relevant attributes, we also need to know what optimization direction you prefer when searching for an apartment.

18. Do you prefer ...

	low	high
... a low or a high amount of rent?	☒	☐
... a low or a high living area?	☒	☐
... a low or a high number of rooms?	☐	☒
... a low or a high floor the apartment is on?	☒	☐
... a low or a high year of construction?	☐	☒
... a low or a high number of parking spaces?	☐	☒
... a low or a high distance to the nearest supermarket?	☒	☐
... a low or a high distance to the nearest public transport station?	☒	☐

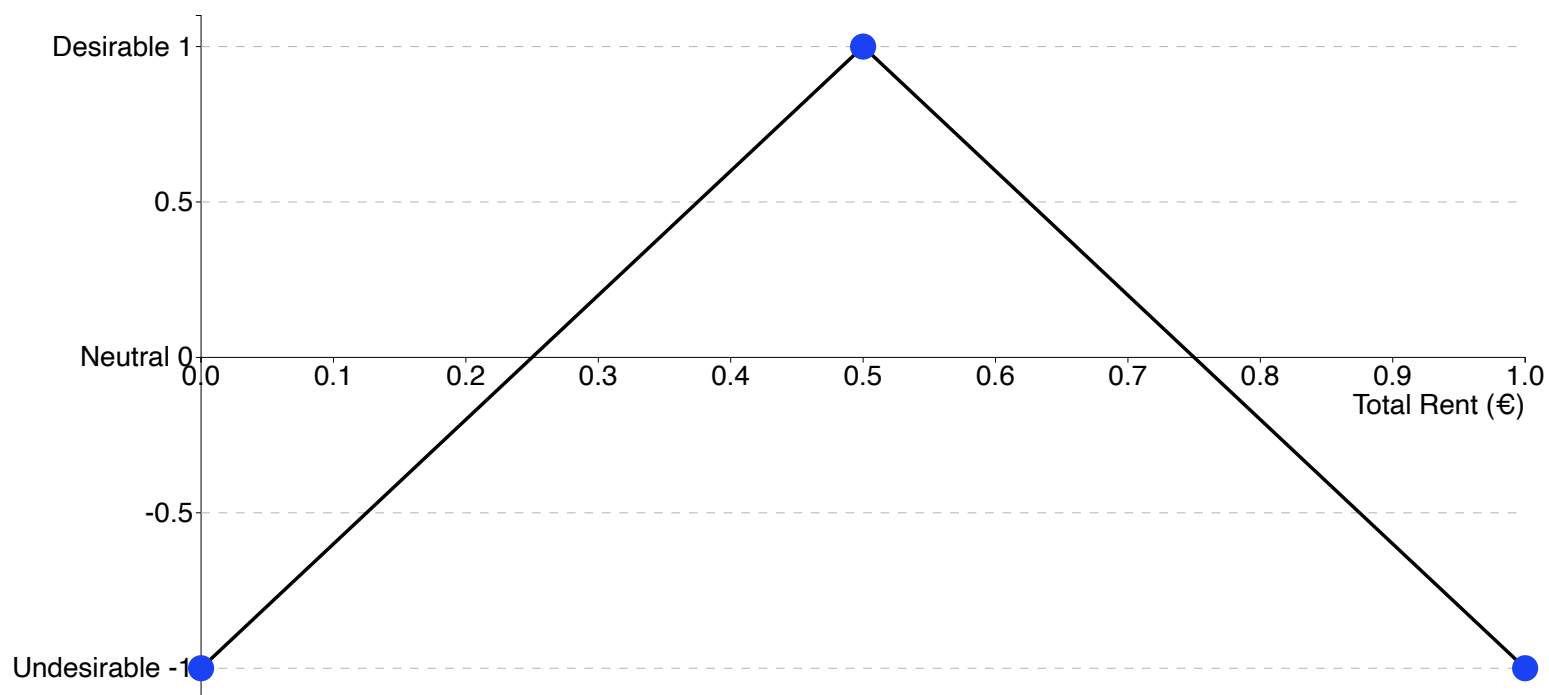
Medium Expressiveness Condition

19. Which amounts of rent do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

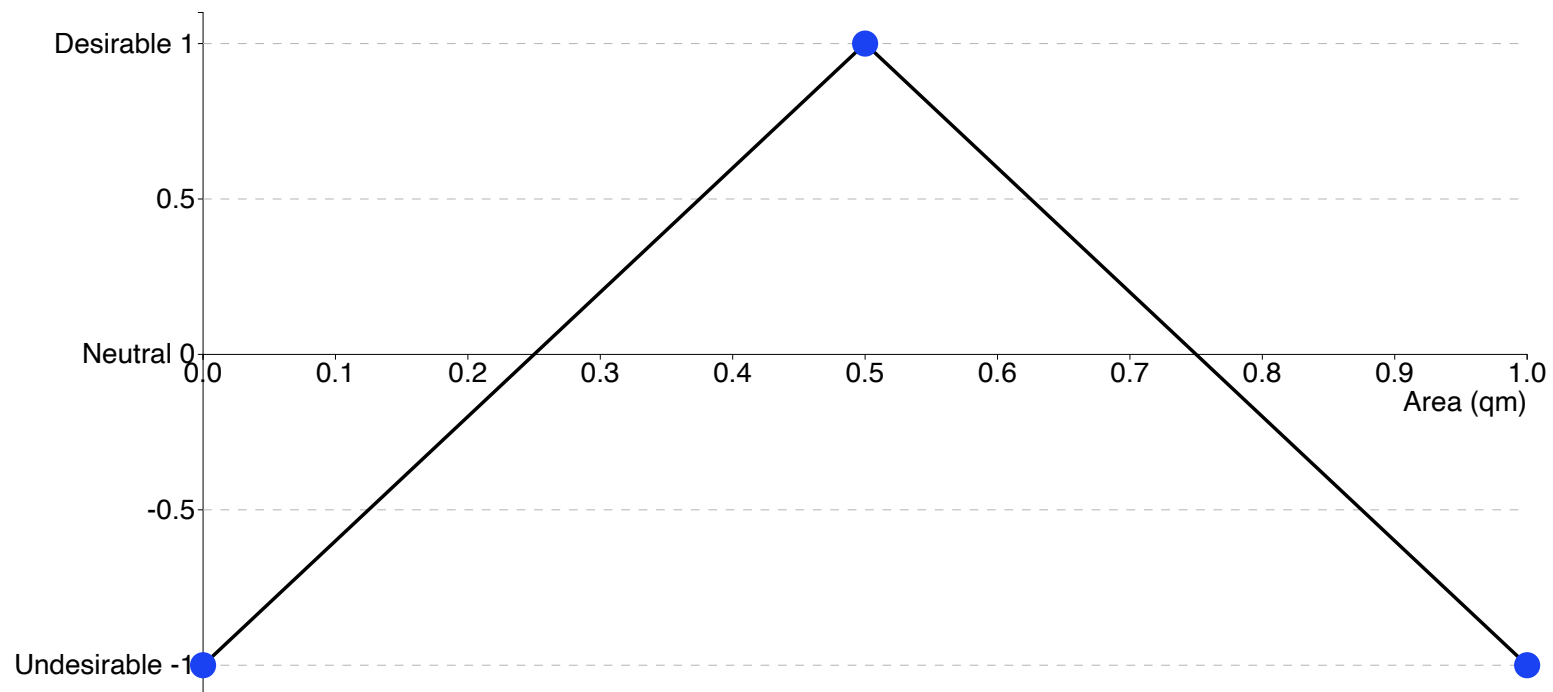


20. Which living areas do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

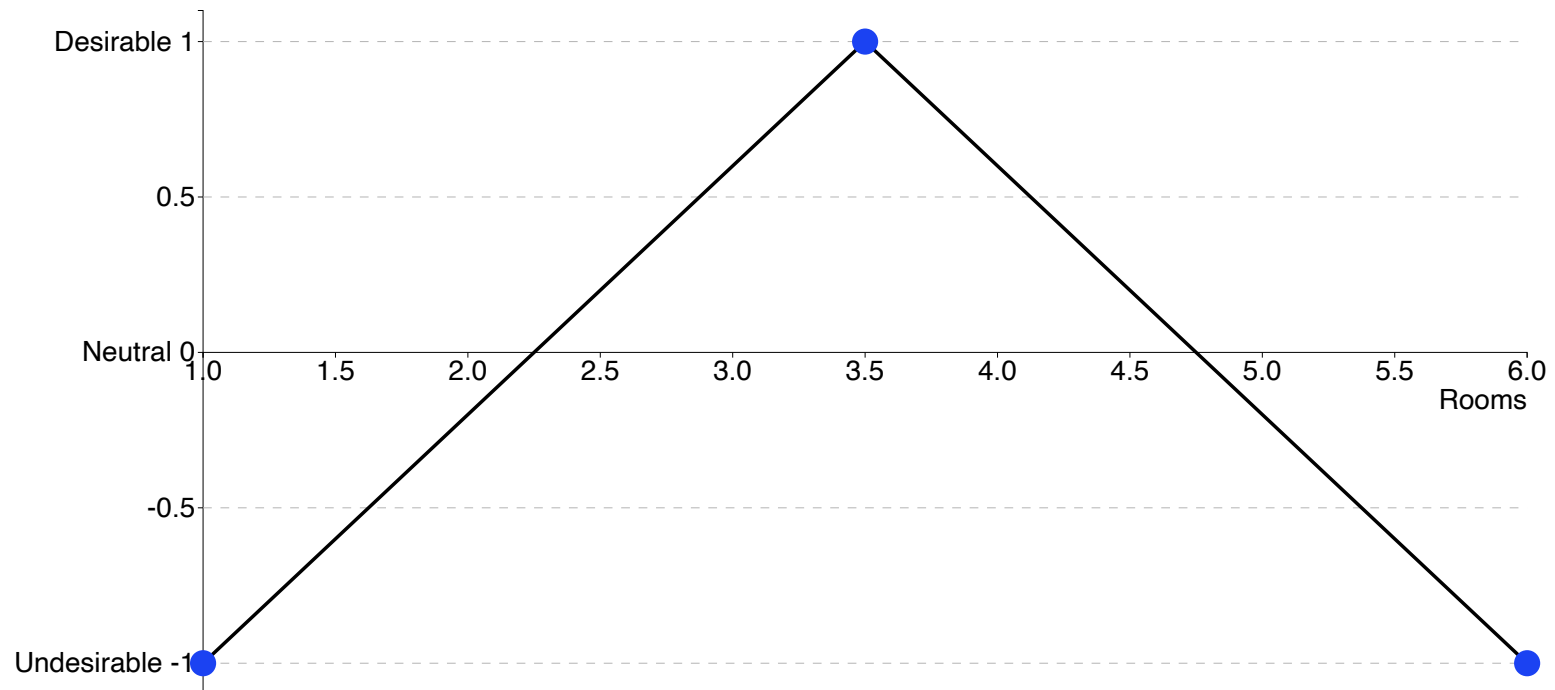


21. Which numbers of rooms do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

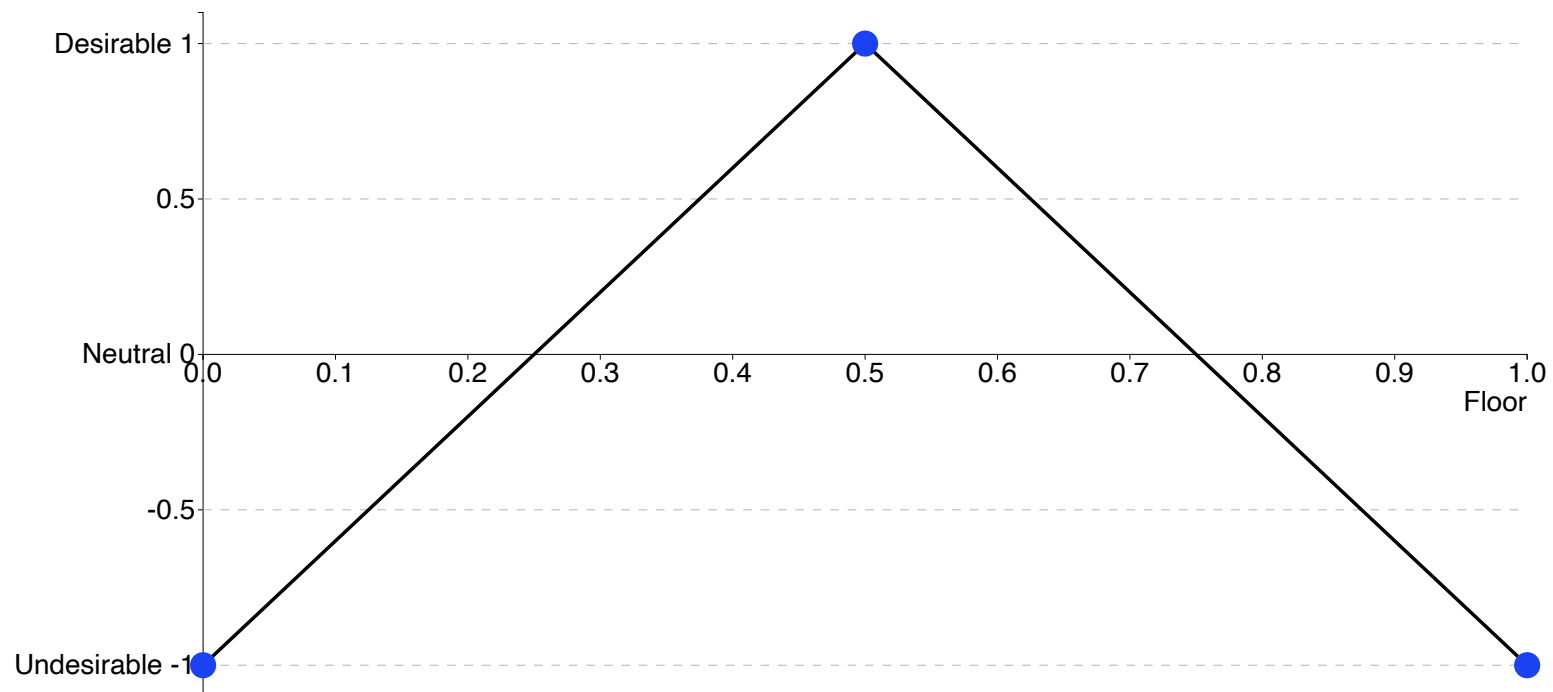


21. Which floors do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

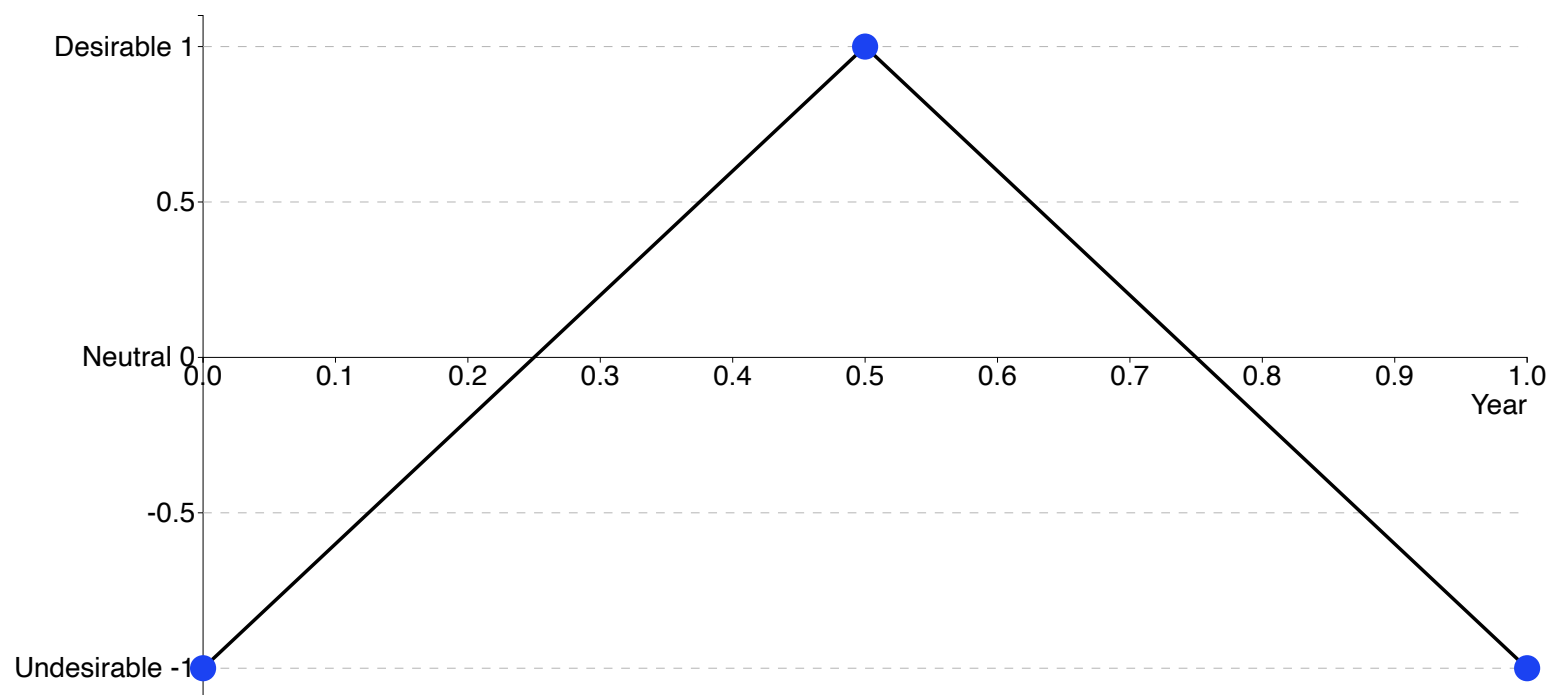


22. Which years of construction do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

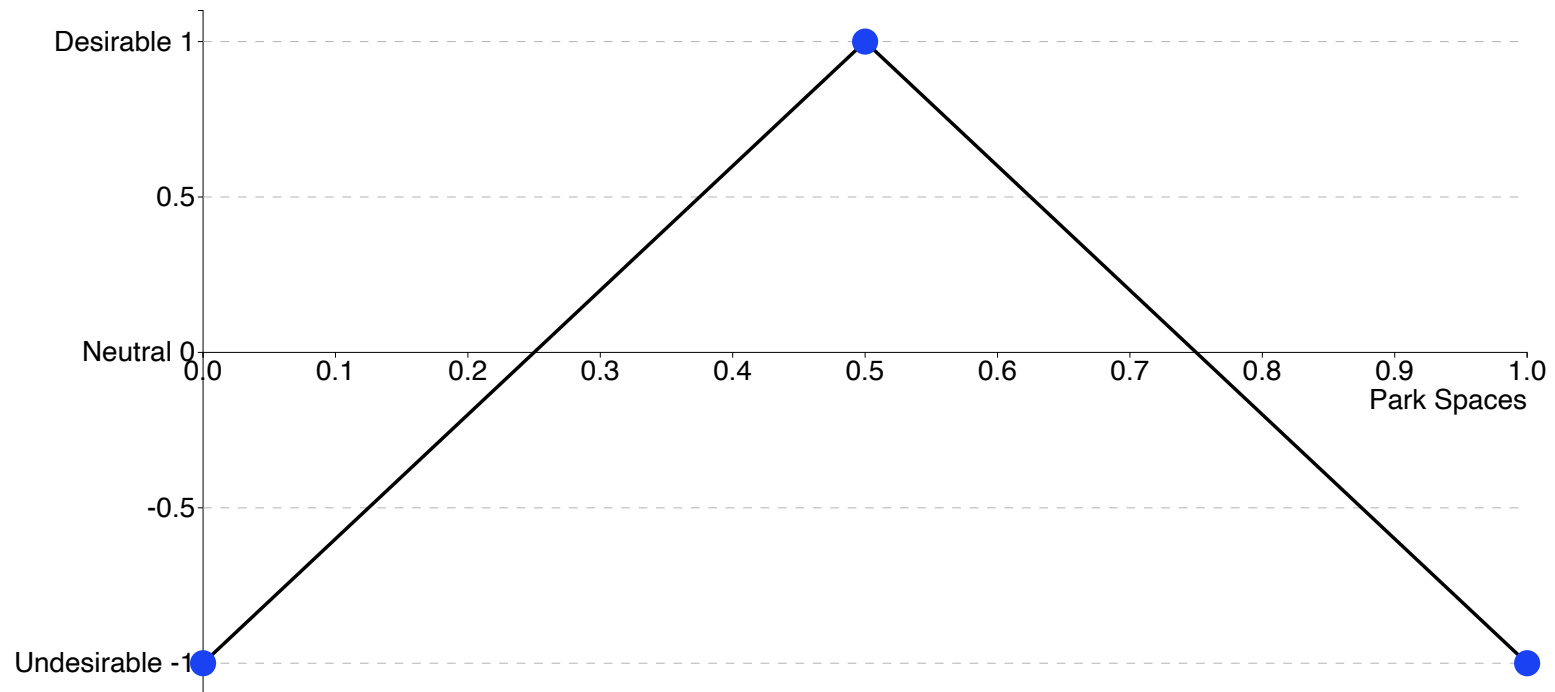


23. Which numbers of parking spaces do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

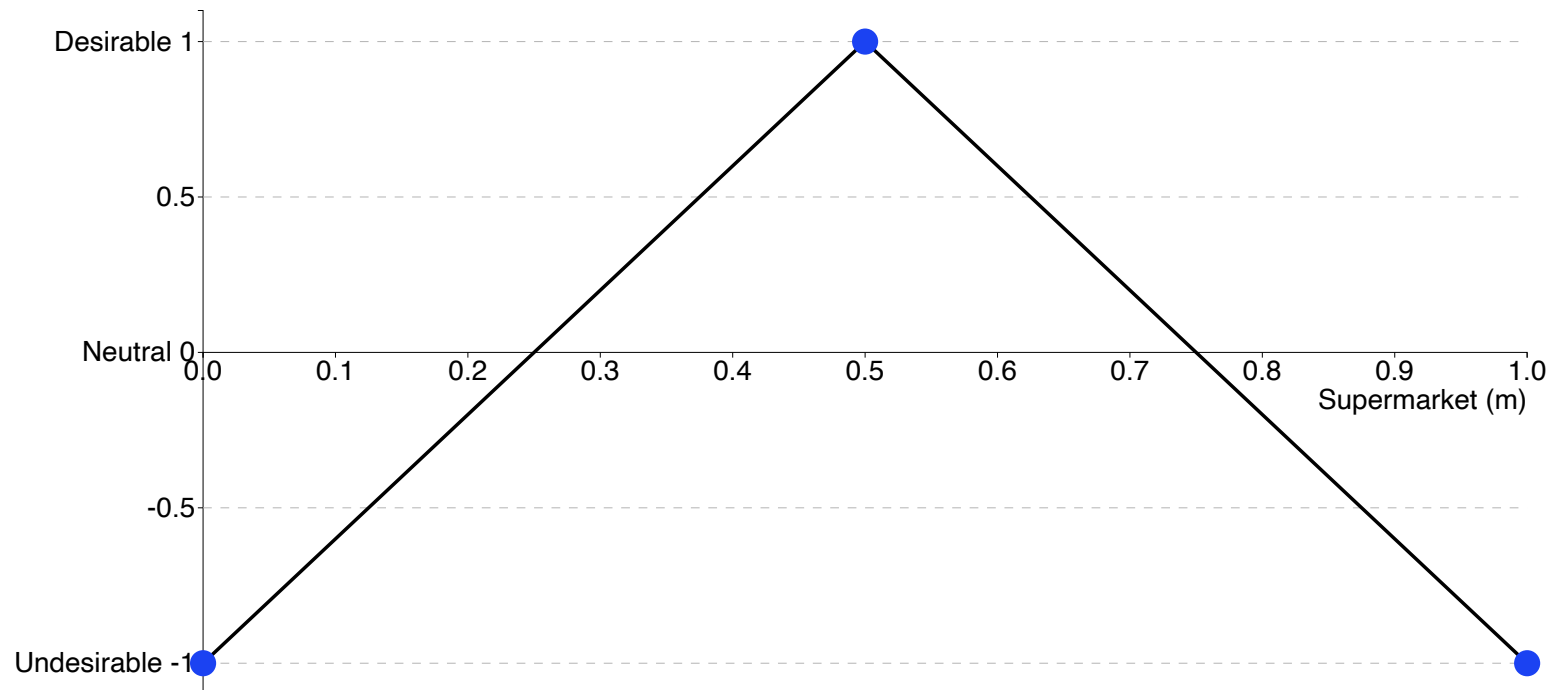


24. Which distances to the nearest supermarket do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.

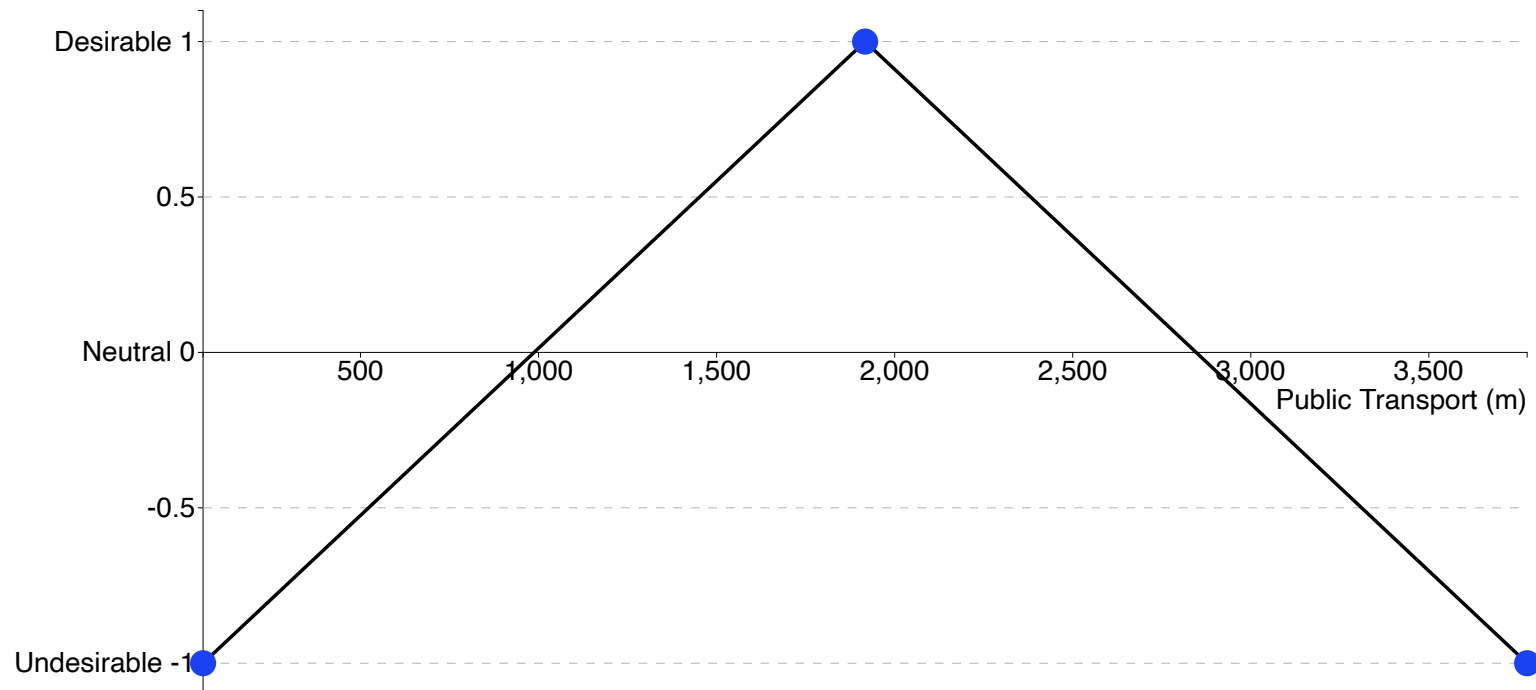


25. Which distances to the nearest public transport station do you prefer and how much?

↕ Flip ↻ Start over Drag points to move them, hover line for tooltip

Warning

Initial curve. Please move points to refine the curve.



High Expressiveness Condition

19. Which amounts of rent do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



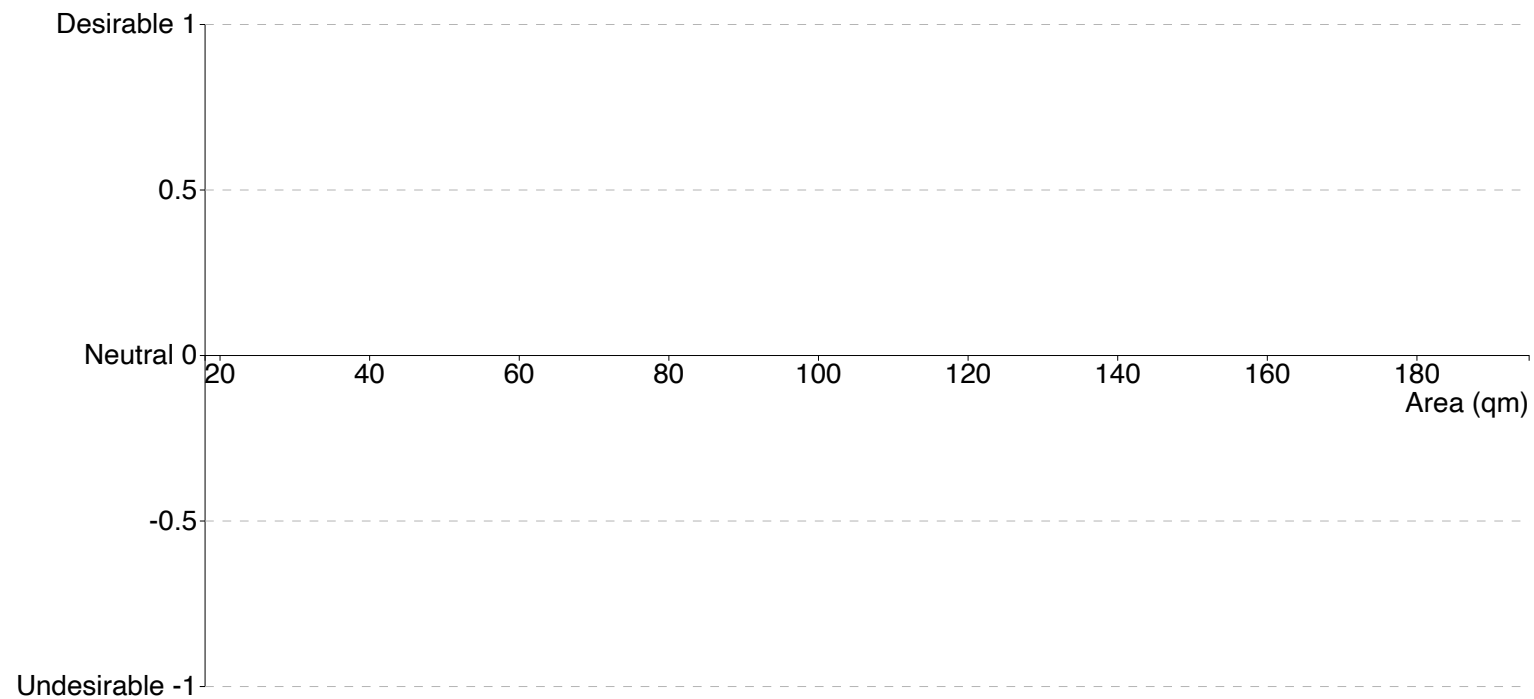
20. Which living areas do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



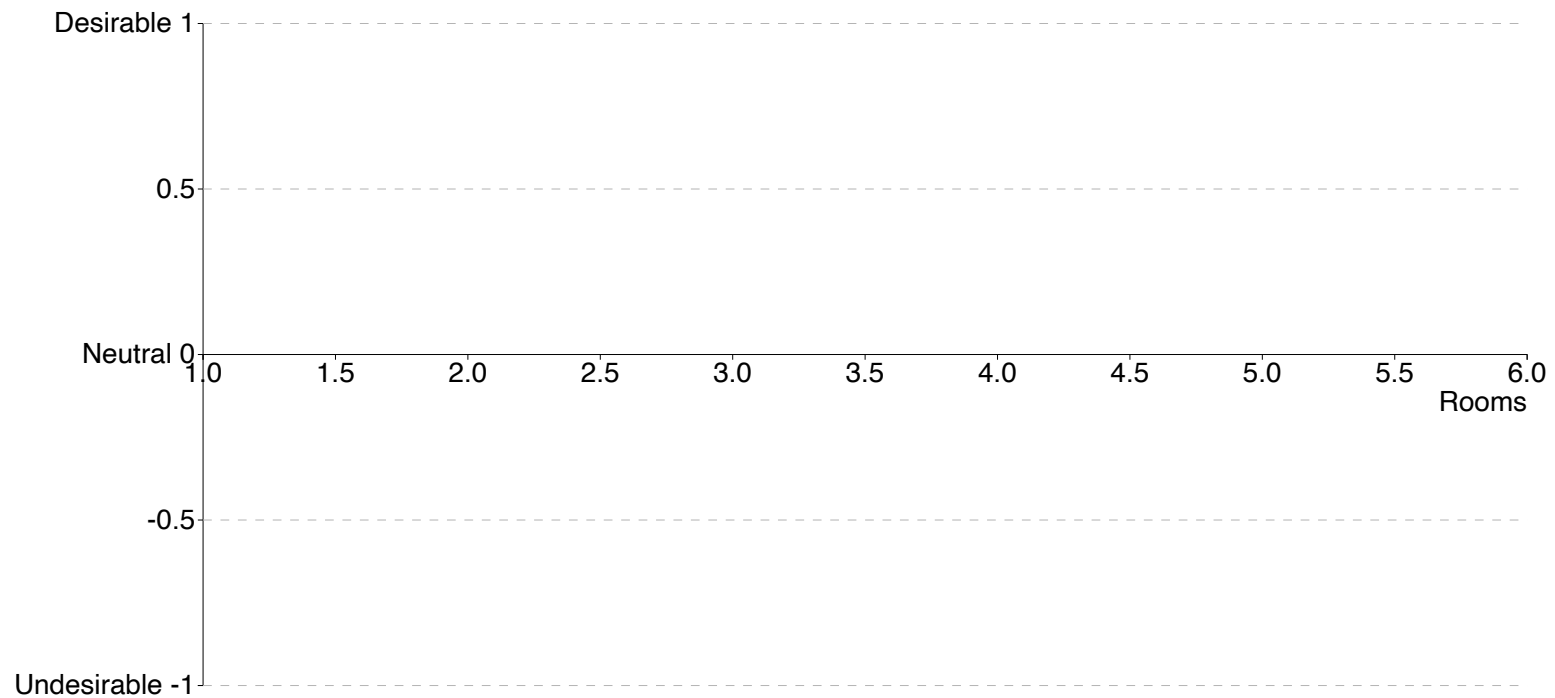
21. Which numbers of rooms do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



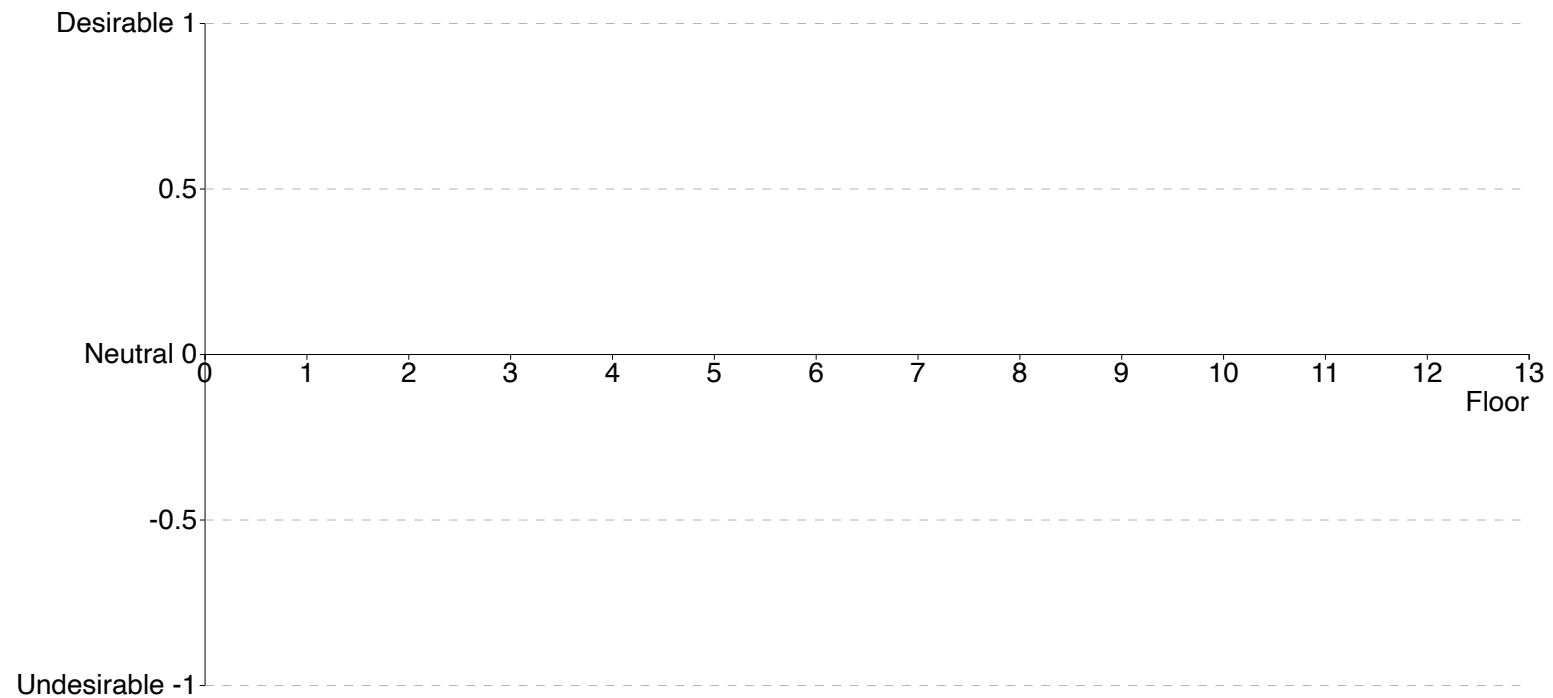
22. Which floors do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



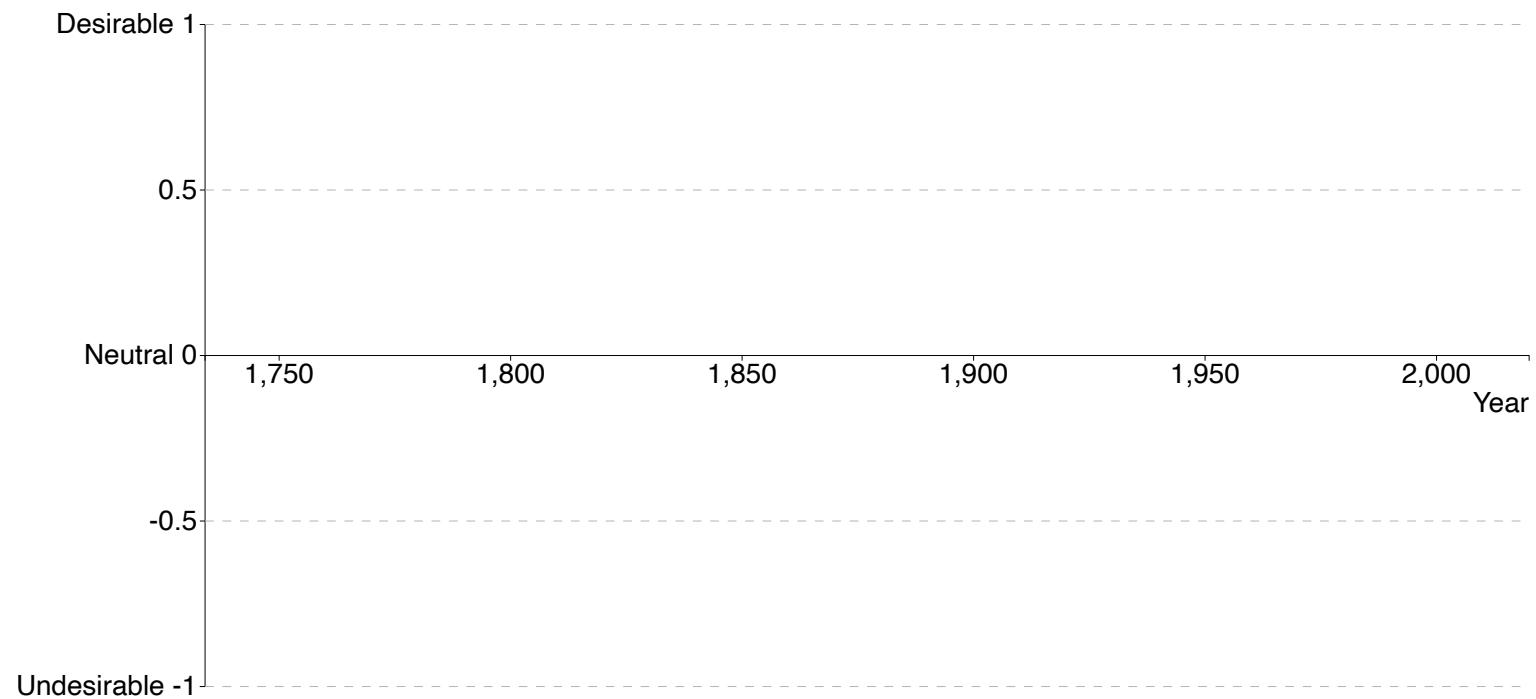
23. Which years of construction do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



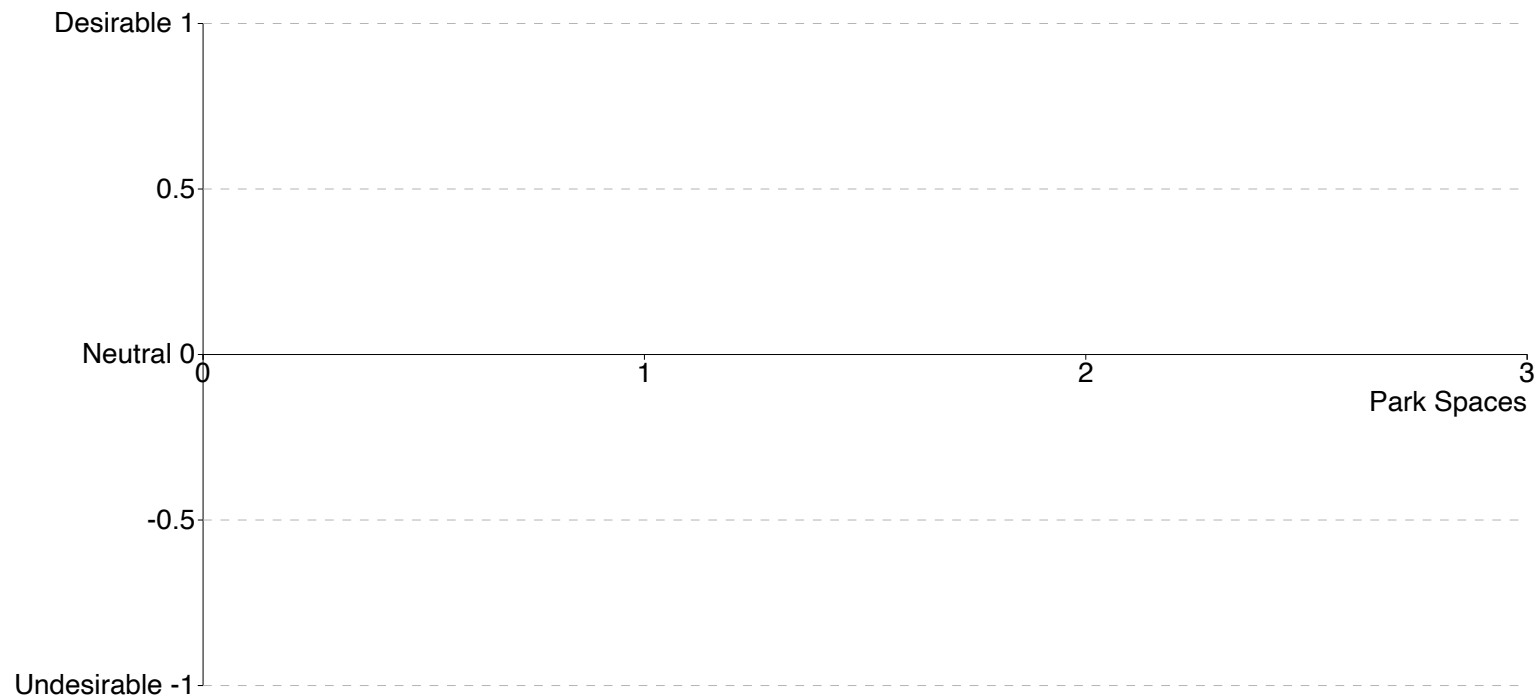
24. Which numbers of parking spaces do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



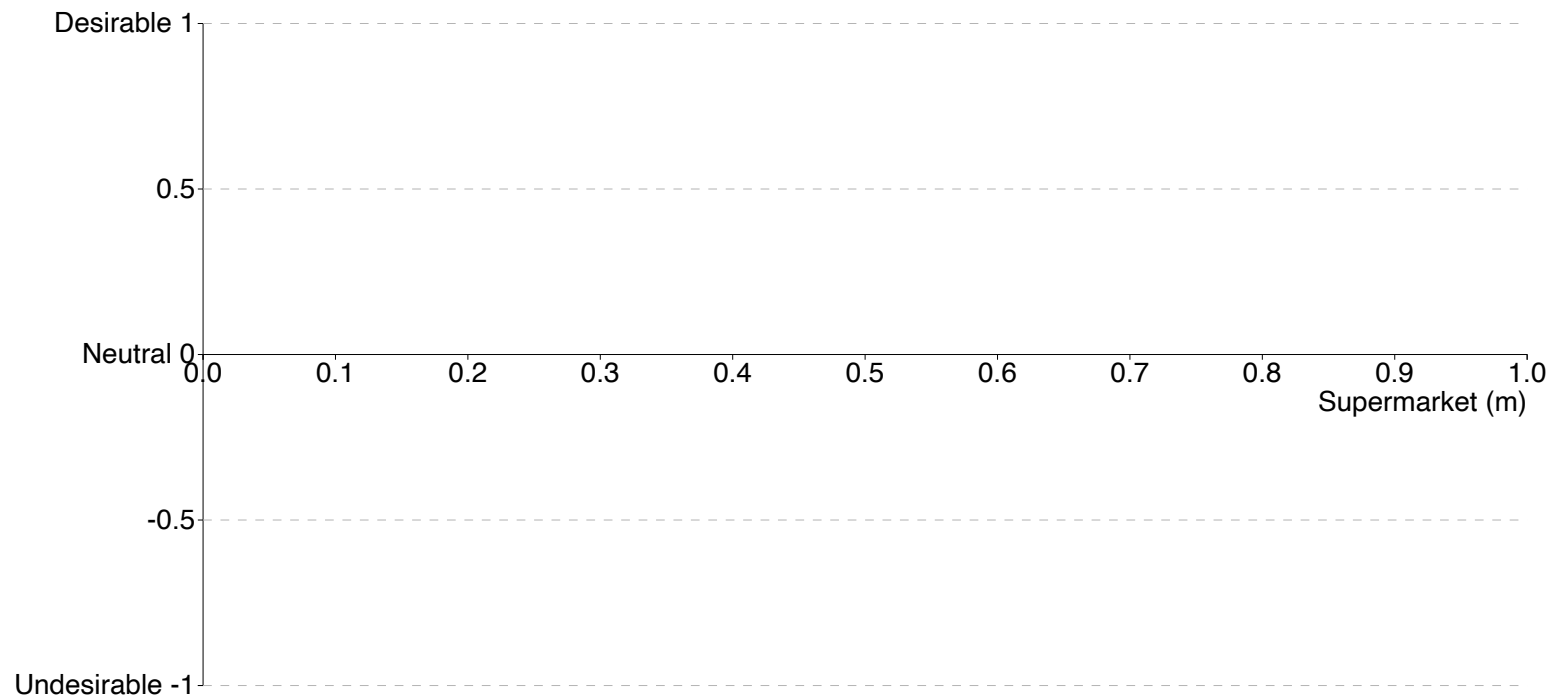
25. Which distances to the nearest supermarket do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



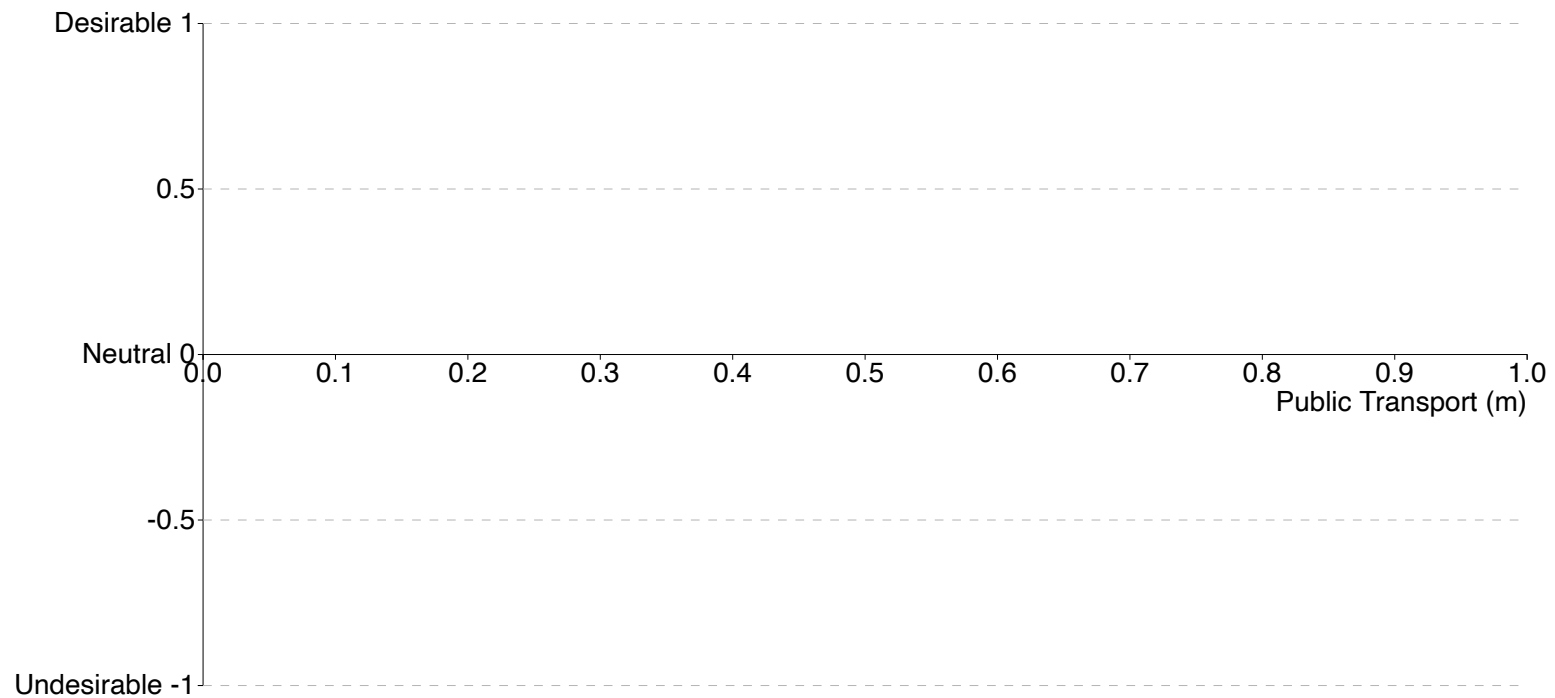
26. Which distances to the nearest public transport station do you prefer and how much?

↕ Flip ↻ Start over

Click anywhere to add a point, drag any point to move it, click a point to select it and hit Backspace to remove it, hover curve for tooltip.

Invalid response

Your curve is empty. Please add points to the curve.



Branching End

Decision Task

On the next pages, you will use the two visualizations to interactively explore different sets of apartments and make your decisions among each set. After using each visualization, you will briefly reflect on the choice you made with that visualization.

24. Which apartment do you choose to rent?

Apartment

Which one would you want to avoid?

Apartment

Apartment	Year	Park Spaces	Area (m ²)	Rooms	Floor	Supermarket (m)	Public Transport (m)	Total Rent (€)
Abucot	2007	2	151	5	2	1842	3316	1171.5
Alibuag	1994	2	98	3	1	193	1653	1200
Alisoso	2013	1	144.02	3	3	3454	1318	1520
Amot	2018	1	61.47	2	0	989	2904	1235.54
Anguapan	2018	1	84	2	1	3839	439	1240
Apo	1910	1	60.64	2	2	2553	821	615
Basan	1984	1	156	5	1	3216	2872	3050
Balaorte	1998	1	76	3.5	3	3564	388	780
Barung	2018	1	64.53	2	3	2671	1614	1294
Bijia	1978	1	97	3	1	2905	1240	1020
Binoalan	2018	2	101.5	3	0	1240	1853	1373
Binuling	2000	1	46	1	0	3137	542	385
Bololo	1903	1	180	3	1	2830	1951	3100
Bulaloc	1966	1	78	3	2	3692	2794	900
Butason	1995	1	61.19	2	2	3228	1822	558
Cabalao	2018	1	122.44	3.5	1	427	474	1695
Caccag	2016	1	82	2.5	10	2154	2325	1820
Calalas	1933	1	74.3	3	3	3173	1269	820
Canawa	1900	1	134.08	5	1	2242	835	1206
Canningag	1936	1	93.5	3	4	1259	2651	703
Casugad	1972	1	63	2	1	238	2608	695
Catongi	1996	1	89.94	3	0	397	108	824.61
Dalapuan	1991	1	93	3	2	1334	1996	910
Dibungco	1982	2	100	4.5	2	1043	535	1250
Gaboc	1905	1	99.81	4	1	3565	1415	1237.73
Higpos	2018	1	126.45	5	1	1797	58	1230
Himamara	2011	1	71	3	0	3813	101	1270
Ikdan	1995	1	72.09	2.5	4	3577	976	595
Inguan	1992	1	24	1	0	987	3153	364
Iwahig	1960	1	34.84	1	2	2953	683	295
Kalakan	2015	1	101	4.5	2	449	1436	1475
Kasile	1924	0	66	3	0	888	1389	503
Kohec	1989	1	80	3	2	673	1757	1050
Lauigan	1995	1	28.5	1	2	4006	550	325
Lolumpan	1954	0	63.7	2	2	232	2577	620
Macapso	1968	1	82	3.5	1	328	3776	835
Majon	2013	2	123.14	4	2	2888	1795	1599
Malanao	2016	1	85	3	2	2699	580	580
Managase	2017	2	105.01	3	0	1055	704	1100
Manalec	2018	0	83	3	3	2641	602	1182
Mandile	1992	1	50	1.5	2	3926	465	550
Manyapit	1927	1	80	3.5	1	2770	1077	600
Mapno	2007	1	90	3.5	4	2084	556	1250
Marcote	2018	1	73	2	5	242	2299	936
Milakeg	1900	2	97.5	3	1	1271	3073	1324
Mulaog	2019	1	63.5	2	1	1657	770	1176
Naruntan	2019	1	90.29	3	1	1027	267	1040
Novele	2016	1	63.04	2	2	2289	2244	814
Nuangan	1972	0	53	3	3	345	1946	577
Oiang	2017	1	195	4.5	3	1619	2520	3400
Pange	2016	2	121	3	13	1809	2259	2560
Pansog	2018	1	88.6	3.5	1	3079	652	1220
Piddan	2018	1	99.82	2	1	1510	1919	1170
Pilahan	2014	1	95.14	3	0	1245	2015	1338
Piluro	2019	1	67	2	1	1487	3539	1400
Sadaan	1956	1	29	1	1	1463	1215	370
Sagneb	2000	2	90	3	0	3027	3695	870
Salito	1900	1	112	3	1	1491	2977	1792
Samak	1993	1	30.38	1	2	2695	1480	290
Sapid	1882	1	97	2	1	3860	2548	840
Sibarok	2017	1	170.3	4	2	2576	476	2141
Silion	1900	1	175	3.5	1	1806	1686	4300
Soom	1978	2	140	5	2	3312	893	750
Sugangon	2019	1	60.98	2	4	1468	661	835.43
Sumbao	1995	1	29.81	1	3	2763	2903	328.5
Tagabili	1962	1	72.21	2.5	3	3096	2585	490
Taguiled	2018	1	49	1	0	2239	1162	800
Talaitan	1903	1	74.59	3	0	1993	2516	494
Talboc	1971	1	63	2.5	1	924	2827	585
Taluay	1989	1	28	1	1	171	2127	450
Tandayag	1996	1	83.5	3	3	3481	3019	610
Tapa	1970	1	49	2	2	685	592	375
Tapon	2018	1	103.35	3	1	2031	2540	1350
Tiji	1970	1	42	1	3	2962	1966	440
Tinagum	2001	1	52	2	0	465	1476	500
Tobog	1968	1	129	4	0	599	711	1064.3
Tombo	1994	1	97	3	2	969	2793	950
Tulayan	1970	1	133	4	2	2289	567	1880
Ulugan	1900	1	85.86	3	2	531	846	460
Vigaan	2018	1	1871	5.5	4	2928	364	3260

The following questions refer to your choice regarding the apartment that you would want to rent.

25. To what extent are you satisfied with your choice?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐

Not at all satisfied Extremely satisfied

26. To what extent are you confident about your choice?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐

Not at all confident Extremely confident

27. To what extent was your choice easy to make?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐ ☐

Extremely difficult Extremely easy

28. Individual preferences and knowledge can significantly impact the decision process. Please respond with “Somewhat agree” to the following statement: “Decisions are highly subjective.”

☐	☐	☐	☒	☐	☐
Strongly disagree	Disagree	Somewhat disagree	Somewhat agree	Agree	Strongly agree

29. Imagine that an automatic recommender system could suggest another choice from the data set taking into account your preferences. Would you switch to this choice?

☐	☐	☐	☐	☐	☒	☐	☐	☐	☐	☐
I would definitely switch										I would definitely stick to my initial choice

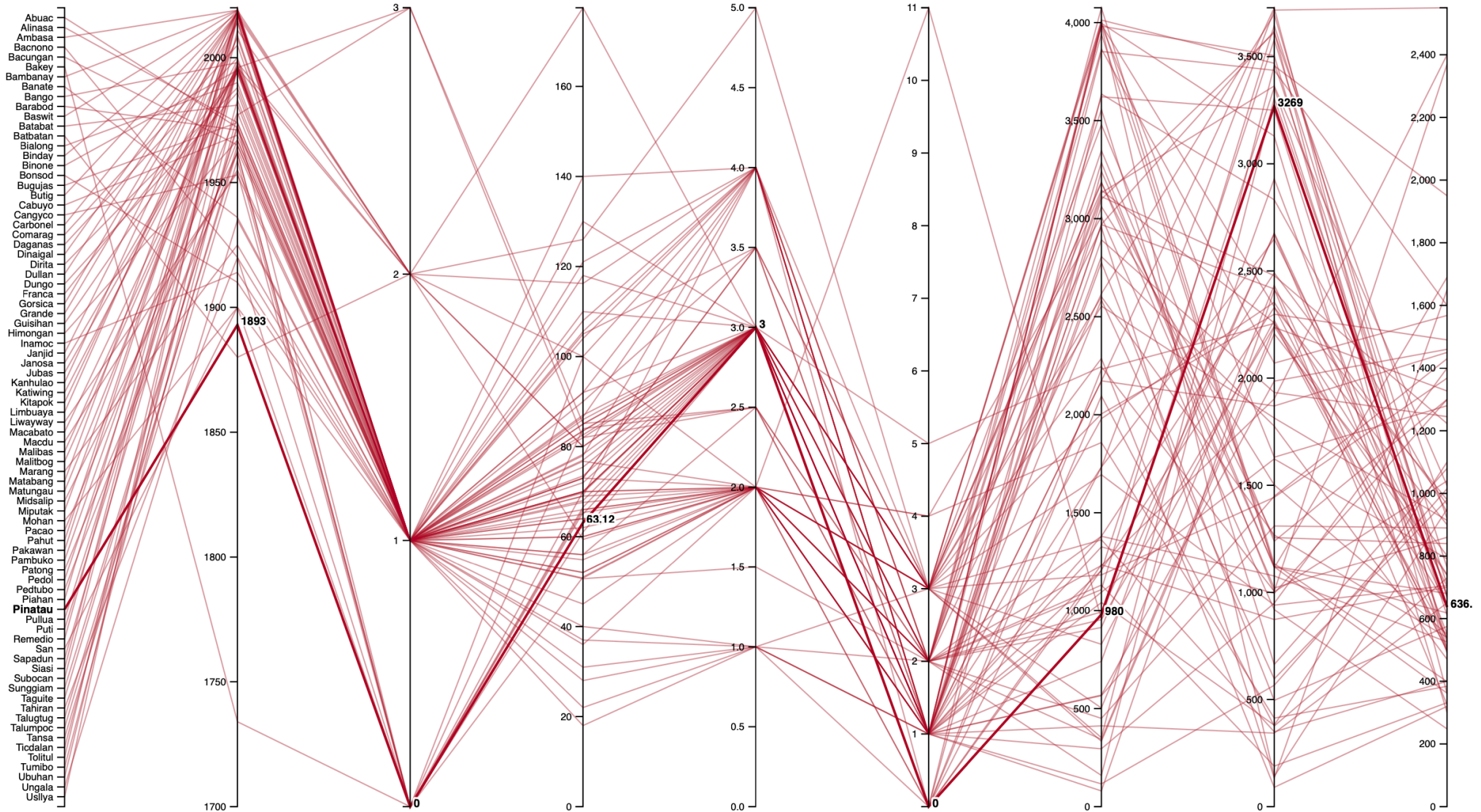
30. Which apartment do you choose to rent?

Apartment

Which one would you want to avoid?

Apartment

Apartment ⇄



The following questions refer to your choice regarding the apartment that you would want to rent.

31. To what extent are you satisfied with your choice?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐ ☐

Not at all satisfied Extremely satisfied

32. To what extent are you confident about your choice?

☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐ ☐ ☐

Not at all confident Extremely confident

33. Please indicate your agreement with the following statement: "I swim across the Atlantic Ocean to get to work every day."

☒ ☐ ☐ ☐ ☐ ☐

Strongly disagree Disagree Somewhat disagree Somewhat agree Agree Strongly agree

34. To what extent was your choice easy to make?

☐ ☐ ☒ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

Extremely difficult Extremely easy

35. Imagine that an automatic recommender system could suggest another choice from the data set taking into account your preferences. Would you switch to this choice?

☐ ☐ ☐ ☒ ☐ ☐ ☐ ☐ ☐ ☐ ☐

I would definitely switch I would definitely stick to my initial choice

Decision Tasks: Success!

You have successfully completed the decision tasks!

We will now ask you about your general experience with the decision-making.

36. How much mental effort have you invested in specifying your preferences?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐

Very low mental effort

Extremely high mental effort

37. How accurately could you express your preferences?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒ ☐ ☐ ☐

Not accurate at all

Perfectly accurate

The following questions refer to your choice regarding the apartment that you would want to rent.

38. How helpful did you find the parallel coordinates to solve the decision task?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☒

Not at all helpful Extremely helpful

39. How helpful did you find the tabular visualization to solve the decision task?

☒ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

Not at all helpful Extremely helpful

40. Please justify your visualization ratings above.

The crossing lines on the second visualisation was very confusing to look at. The table was also very difficult to interact with because everything is on top of each other and indistinguishable from each other. The first visualisation was simple because I could construct my own graphs and decide from there, I did not have to try and decipher a complicated table first.

Nearly Done!

You are nearly done with the questionnaire!

We just need some final information concerning your person.

41. What gender do you identify most with?

Male ▼

42. What is your age?

20 to 29 years ▼

43. What is the highest degree or level of education that you have completed?

Bachelor's degree ▼

44. How would you describe your personal level of decisiveness?



Very indecisive



Very decisive

45. How would you describe your experience with using or developing data visualizations?

☐

No experience

☐☐☐☒

Very experienced

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Thank you for completing this questionnaire!

We would like to thank you very much for helping us.

By clicking "Submit", your answers will be transmitted and you will be redirected to the Prolific page.

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redirComp

Notice

Redirect → <https://app.prolific.com/submissions/complete?cc=CDXUMJP5>